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Permalink

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Journal

Journal of Personality and Social Psychology, 126(3)

ISSN

0022-3514

Authors

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Cheng, Rongxin
Robins, Richard W

Publication Date

2024-03-01

DOI

10.1037/pspp0000489

Peer reviewed

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AMERICAN PSYCHOLOGICAL ASSOCIATION

Journal of Personality and Social Psychology:
Personality Processes and Individual Differences

© 2023 American Psychological Association
ISSN: 0022-3514

2024, Vol. 126, No. 3, 492–510
https://doi.org/10.1037/pspp0000489

Investigating the Trajectory of Mexican-Origin Youth's Materialistic Values Across Adolescence and Prospective Associations With Life Satisfaction in Early Adulthood: The Role of Familism, Demographic Factors, and Parent Characteristics

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Using data from a 14-year longitudinal study of 674 Mexican-origin youth, the present study examined the development of materialistic values (MV) across adolescence (ages 10–16), correlates of the adolescent MV trajectory, and prospective associations between the MV trajectory and life satisfaction in early adulthood (ages 17–23). Latent growth curve analyses showed that MV decreased, on average, from early to late adolescence. Youth raised in families with higher household income, higher maternal and paternal education, and lower maternal and paternal MV tended to be less materialistic; higher maternal education was associated with smaller decreases in MV from age 10 to 16. A strong association of greater decreases in MV across adolescence was found with higher initial life satisfaction at age 17, but not with changes in life satisfaction from age 17 to 23. Findings from this study provide insights into the development of MV among Mexican-origin youth.

Keywords: materialistic values, life satisfaction, familism, gender, maternal/paternal education

Supplemental materials: https://doi.org/10.1037/pspp0000489.supp

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Materialism describes "the importance a person places on possessions and their acquisition as a necessary or desirable form of conduct to reach desired end states" (Richins & Dawson, 1992, p. 307). It is characterized by a set of materialistic values (MV) that are learned and subsequently guide one's daily life.

In the literature, individuals with higher levels of materialism are found to display more psychobehavioral maladaptation. For example, people with higher levels of materialism are poorer at money management (Donnelly et al., 2012) and have more favorable attitudes toward borrowing money, overspending, and becoming indebted (e.g., de Matos et al., 2019; Watson, 2003). More materialistic people are more likely to engage in maladaptive consumption, including impulsive (e.g., Richins, 2011; Yoon & Kim, 2016) and compulsive purchases (e.g., Islam et al., 2017; Ruvio et al., 2014), conspicuous and social status-driven purchases (e.g., Podoshen & Andrzejewski, 2012; Shi et al., 2016). When MV are more strongly held and override other values, people are more likely to lack a sense of security (Burroughs & Rindfleisch, 2002), express more self-doubts (Chang & Arkin, 2002), experience higher social anxiety (Schroeder & Dugal, 1995), and have lower self-esteem (e.g., Park & John, 2011; Ruvio et al., 2014), especially self-esteem that is contingent on intrinsic factors such as being true to oneself and personal growth (Nagpaul & Pang, 2017). More materialistic individuals are less satisfied with their lives (e.g., Baker et al., 2013; Barnfield, 2020), reporting lower subjective well-being (e.g., Bauer et al., 2012; Dittmar et al., 2014). Adolescents, in particular, with higher levels of materialism are found to exhibit more externalizing behavior symptoms, such as attention-deficit disorder and conduct disorder symptoms (Cohen & Cohen, 1996).

This article was published Online First November 13, 2023.
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This research was supported by a grant from the National Institute on Drug Abuse (Grant DA017902) to Richard W. Robins and a grant from the National Natural Science Foundation of China (Grant 71872189) to Bing Shi. The authors thank the participating families, staff, and research assistants who took part in this study. Several other articles have used data from the California Families Project. However, none of these studies examined materialistic values and their impact on life satisfaction.

Study materials and R scripts are available on the Open Science Framework at https://osf.io/nf8zj/?view_only=d444117406f944e987a2644f265ede41. The authors posted the anonymized data for this study except the demographic variables (e.g., gender, household income, parental education level). Researchers interested in replicating findings can also contact Richard W. Robins to gain access to individual-level data. The hypotheses and analysis plan were not preregistered.

Bing Shi played a lead role in conceptualization, investigation, and writing–original draft and a supporting role in data curation, formal analysis, funding acquisition, and methodology. Rongxin Cheng played a lead role in data curation, formal analysis, and methodology and a supporting role in investigation and writing–original draft. Richard W. Robins played a lead role in funding acquisition, project administration, resources, and supervision and a supporting role in conceptualization, formal analysis, investigation, methodology, and writing–original draft.

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As concerns grow over rising levels of materialism among youth (Barnfield, 2020; Schor, 2004), it is crucial to understand the development of MV and the prospective associations of MV with life satisfaction and other developmental outcomes. However, the vast majority of studies on materialism are still based on concurrent associations in adult samples (e.g., Grougiou & Moschis, 2015; Islam et al., 2017), and only a few studies have examined adolescent MV (e.g., Barnfield, 2020; Cohen & Cohen, 1996; Flurry & Swimberghe, 2021; Shi et al., 2016); as Dittmar et al. (2014) state, "there is a paucity of studies of materialism with younger samples" (p. 914). Note that adolescence is a critical stage when youth form fundamental values, beliefs, and behavioral patterns that are consequential in adulthood. The present study aims to examine the developmental trajectory of MV across adolescence and its prospective associations with life satisfaction in early adulthood. Moreover, in studies with cross-sectional design, there often exists a primary problem of "the age by cohort confound" (Sankoff, 2008). The seriousness of this problem relates to the dependent variable. For example, the less "basic" or "biological" the dependent variable, the more likely it is that "the age by cohort confound" will be present (Miller, 1998). Considering that MV are learned through everyday interactions and the widespread of the age span, longitudinal methodology is adopted in this study.

We focus on Mexican-origin adolescents because Latinxs are a large and rapidly growing ethnic group in the United States. Hitherto, this ethnic group continues to experience lower socioeconomic status (SES) than Americans of other ethnic backgrounds such as African Americans and European Americans (Barger et al., 2009). The lower SES may increase the risk of materialism endorsement among Mexican-origin youth, considering that youth growing up in poor families place more importance on financial success and wealth (e.g., Cohen & Cohen, 1996; Ku, 2015), showing a tendency of being more materialistic than their affluent counterparts (Chaplin et al., 2014).

Furthermore, Knight et al. (2011) have developed the Mexican American Cultural Values Scale (MACVS), assessing a set of contemporary mainstream American values (including MV) and a set of Mexican American traditional values of which familism is a central traditional value for Mexican-origin youth in the United States. Familism involves several commonly recognized dimensions, including family support and cohesion (relying on family for support when problems arise and valuing warm relationships), family obligations (responsibility to nuclear and extended family to provide material and emotional support), and family as a referent (understanding that one's behavior reflects upon the family; Knight et al., 2011; Sabogal et al., 1987; G. L. Stein et al., 2014). Research has documented the importance of familism for promoting positive family relationships and health behaviors and reducing mental health problems and risk behaviors (e.g., G. L. Stein et al., 2014; Wheeler et al., 2017). Considering that familism is a source of social support, social connection, and social reference that is especially valued by Latinx individuals (e.g., Padilla-Walker et al., 2020), we examine it as a factor that might develop in connection with the development of MV among Mexican-origin adolescents and, furthermore, might moderate prospective associations of the MV trajectory across adolescence with young adult life satisfaction.

In summary, the present study aims to (1) identify the normative trajectory of MV and the codevelopment of MV and familism across adolescence (ages 10–16); (2) identify antecedent factors that predict the MV trajectory across adolescence, including youth gender and nativity, household income, parent MV, familism, and education level; (3) investigate prospective associations of the adolescent MV trajectory with life satisfaction in early adulthood and the potential moderating effects of youth familism, gender, nativity, and household income. This study is one of the only longitudinal studies of the development of MV across adolescence and one of the few studies to examine the antecedents and consequences of the adolescent MV trajectory. Findings should contribute to the literature by showing how MV change across adolescence, how the MV trajectory associates with life satisfaction in early adulthood, and how the development of familism (a core Mexican traditional value) is associated with MV (a core American value).

Social Identity Development and Integration Within the Self: A Neo-Piagetian Perspective

While classic intergroup theories have focused primarily on the situational and short-term changes in social identities (Tajfel, 1981), recent research has begun to explore the developmental changes in social identities and their integration within the self (Amiot et al., 2007). To address this issue, social identity is defined as "that part of the individual's self-concept which derives from his or her knowledge of membership to a social group (or groups) together with the value and the emotional significance attached to it" (Tajfel, 1981, p. 255).

To account for the dynamic and complex processes by which social identities develop and become integrated within the self over time, we adopt a neo-Piagetian developmental approach. This approach is particularly appropriate, as it proposes that the self develops from fractionation and differentiation of its facets toward integration within the self, which becomes increasingly complex over time (e.g., Fischer, 1980; Harter, 1999; Yan & Fischer, 2002). Mexican-origin youth need to develop at least two salient social identities through their everyday interactions: American citizen identity and Hispanic ethnic identity. The development of these two identities is posited to undergo different paths before they become organized within the self-structure, while their relative importance to the overall self-concept is maintained. Additionally, this approach recognizes that transitions and significant life changes represent sensitive periods for social identity development and integration within the self (Harter, 1999). Adolescence, accordingly, marks a critical period for Mexican-origin youth's identity development and integration within the self.

More specifically, Grotevant (1987) proposed that the development of each social identity consists of four major components. The first component involves the process of identity development within a specific domain, such as occupation, political ideology, values, and views of oneself in friendships and committed relationships. As we will discuss in more detail in the following section, MV represent a specific domain in adolescents' social identity development among peers. For Mexican-origin youth, the development of MV in adolescence denotes part of the process of their American identity development. The second component pertains to the interdependencies among developments in different domains, as reflected in the connections between identity developmental processes in different domains. Mexican-origin adolescents need to navigate between Hispanic ethnic identity embedded in Mexican traditional

culture and American identity embedded in American mainstream culture. Note that MV reflect contemporary mainstream American values and familism is a central Mexican traditional value for Mexican-origin youth in the United States (e.g., Knight et al., 2011; G. L. Stein et al., 2014). It would be of interest to probe the interdependency between developments in MV and familism among Mexican-origin youth.

Next, the third component involves individual characteristics that can influence the process of identity development. Gender, for example, can have a significant impact on the development of social identities, considering long-acknowledged differences in areas such as gender stereotypes (e.g., Retelsdorf et al., 2015; J. A. Stein et al., 1992) and social interactional patterns (e.g., Benenson, 1993). In addition, some of the Mexican-origin youth have been in the United States for several generations, while others have recently immigrated. Nativity status varies among them and should have the potential to influence their identity development. The fourth component pertains to the contexts of development, including cultural and societal factors, family, school, and work environments. From a neo-Piagetian perspective, support from significant others in the context of development plays an important role in social identity development (e.g., Granott et al., 2002; Harter, 1999). Significant others (e.g., parents) can help individuals develop their social identity through scaffolding and bridging processes, which often involve establishing cognitive connections between preexisting self-components and new materials to be integrated (Fischer, 1980; Granott et al., 2002), pointing out conflicts among identities, and providing support to resolve these conflicts (Harter, 1999). Taking these together, we aim to examine Mexican-origin youth's gender, nativity status, and factors relevant to support from their parents such as parent familism, education levels, and household income as antecedents of the development of MV in adolescence. By considering the four components, we can gain a more nuanced understanding of the complex and multifaceted processes of Mexican-origin youth's social identity development and integration within the self.

MV: A Specific Domain Involved in Adolescents' Social Identity Development

Adolescence is a challenging, developmental stage that is marked by accelerated development in areas such as cognition, self-regulation, and family and peer relationships. As the primary socialization agent shifts from parents to peers during this period (Moschis & Churchill, 1978; Shi & Xie, 2012), creating a social identity among peers and establishing satisfactory peer relationships become particularly salient (Erikson, 1968, 1994; Richins, 2017). Adolescents often turn to shopping and consuming material possessions as a means of achieving this goal and developing a sense of identity among peers (John, 1999). They become increasingly aware of and use the social aspect of consumption activities to satisfy their sociopsychological needs, such as self-expression and self-enhancement among peers. As a result, consumption becomes a self-defining and self-expressive behavior that serves to "produce a desired self through the image and styles conveyed through one's possessions" (Thompson & Hirschman, 1995, p. 151).

Consistently, consumer researchers have long recognized that adolescents are consumers in training (John, 1999; Mangleburg et al., 1997). Chaplin and John (2005), with a focus on the self-concept and brand behavior of children and adolescents, found that adolescents apparently view brands as symbolizing their reference group affiliation and as connecting to their self-concepts. Richins (2011) identified two categories (i.e., transformation of self-identity and transformation of relationships with others) that people expect to gain by making a purchase. While transformation expectations have not been studied among children, in adults the relationship between materialism and transformation expectations is reliably strong (e.g., Donnelly et al., 2016; Richins, 2011). Given the critical developmental task and consumer socialization process among peers, the development of MV in adolescence should be closely involved in their identity development among peers.

Developmental Trajectory of MV Across Adolescence

John's (1999) theory of consumer socialization posits that, during early adolescence (ages 11–15 years), youth show a heightened need to shape their social identity and conform to group expectations, increasing the importance of social comparisons with other peers (e.g., Chan & Prendergast, 2007; Smolak & Stein, 2006). Because of a bias toward making upward social comparisons (Buunk & Gibbons, 2007), comparisons with others can induce a sense of inferiority or deficiency. Youth may attempt to compensate for feelings of inferiority through material acquisition (Reeves et al., 2012), such as purchasing popular clothing or the latest electronic items, to make themselves look "cool" and obtain higher status.

Moreover, Wooten (2006) discovered that adolescents establish consumption norms and values among their peer groups and that violating these norms and values will subject them to negative appraisals (i.e., ridicule). Peer ridicule about products, brands, or other possessions can damage adolescents' self-esteem, which is particularly fragile at the onset of adolescence (Rosenberg, 1986). Children and adolescents also use consumption to bolster their self-esteem, especially as self-esteem declines during early adolescence (around age 12 or 13; Eccles et al., 1989; Twenge & Campbell, 2001). Using cross-sectional data on a sample of 8–18 years old, Chaplin and John (2007) found that materialism is highest in early adolescence and lower in both middle childhood and late adolescence. Children and adolescents' materialistic orientations resemble a strong, age-related pattern of self-esteem. That is, the decline in self-esteem, when it occurs around age 12 or 13, may trigger an interest in material goods, the possessing of which is suddenly perceived as one way to enhance the self. However, according to the same study, as self-esteem rebounds in late adolescence, the need for self-enhancement and materialistic orientation declines. Note that there is a dearth of longitudinal research on the development of MV across adolescence, raising the possibility that the age differences reported by Chaplin and John (2007) were due to cohort differences rather than developmental changes.

Over time, with the emerging mastery of the peer relationship developmental task, the cumulative result of daily peer interactions improves the youth's competence in interpersonal skills, which suggests less reliance on tangible resources or material possessions to maintain one's identity and satisfactory relationships with peers. Based on theoretical models of developmental tasks during the transition into adolescence (e.g., Erikson, 1968; Richins, 2017) and previous cross-sectional findings (e.g., Chaplin & John, 2007; Twenge & Campbell, 2001), we hypothesize that:

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Hypothesis 1 (H1): Endorsement of MV will decrease over time from early to late adolescence.

We focus on Mexican-origin adolescents who navigate between Hispanic ethnic identity embedded in Mexican traditional culture and American identity embedded in American mainstream culture. Creating an identity among American peers in adolescence should set the foundation for the development of their American identity in the society. Note that MV reflect contemporary mainstream American values, familism reflects values associated with Mexican and Mexican American beliefs, behaviors, and traditions (Knight et al., 2011). We aim to explore the interdependency between developments in MV and familism (i.e., the codevelopment of MV and familism) without a specific hypothesis. Simultaneous and longitudinal measurement of these two constructs would be adopted, considering that using longitudinal data allows for more accurate measurement of developmental processes than cross-sectional or other method-relevant data.

Antecedents of the Developmental Trajectory of MV

In addition to the developmental trajectory, we examined several antecedents of MV: youth gender and nativity status, household income, parent MV, familism, and education level.

Youth Characteristics

We first focus on gender differences in the trajectory of MV. Based on survey data for a large national sample of U.S. high school seniors, females are less likely than males to accept materialism and competition, whereas more likely than males to express concern and responsibility for others' well-being (Beutel & Marini, 1995). Studies based on a large sample of British secondary school adolescents (Flouri, 2004) and a sample of Mexican college students (Felix et al., 2013) found that boys are more likely than girls to endorse MV. A number of other studies have reliably demonstrated higher materialism scores for men than for women (e.g., Kamineni, 2005; Segal & Podoshen, 2013).[1]

Regarding nativity status, first-generation Mexican-origin youth (i.e., born in Mexico) are generally less acculturated than second+ generation Mexican-origin youth (i.e., born in the United States) and consequently may be less likely to endorse contemporary mainstream American values such as materialism. However, there is little relevant theory or research that could provide evidence for this pattern. We examine nativity without a specific hypothesis.

Lower income families may be associated with youth's more endorsement of MV than higher income families, considering that teenagers with higher levels of materialism tend to come from families with lower income levels or less advantageous socioeconomic circumstances (e.g., Chaplin et al., 2014; Goldberg et al., 2003). Youth from poor families, when they reach adolescence (ages 11–13 years), are more materialistic than their wealthy counterparts (Chaplin et al., 2014). Children who grow up in poor families (e.g., Cohen & Cohen, 1996; Ku, 2015) place more importance on financial success and wealth than their affluent counterparts. Wooten (2006) has pointed out that peer ridicule as it pertains to brands and consumption can become a serious issue for working-class adolescents, which may be one reason for this segment to value and pursue expensive goods more than upper-class adolescents.

Hitherto, little and even none empirical research is available to make predictions of gender and household income as antecedent factors of MV *changes* over time. In this research, we make predictions based on the reported associations of these two factors with youth's endorsement of MV. Assuming these associations remain consistent over time, we further speculate on the associations of gender and household income with the developmental changes in MV. For example, a considerable number of studies demonstrate stronger endorsement of MV among boys than among girls (e.g., Beutel & Marini, 1995; Felix et al., 2013). We predict higher initial levels of MV among boys than among girls in adolescence. Furthermore, we speculate that boys may exhibit smaller decreases in MV over time compared to girls.

Hypothesis 2 (H2): Adolescent boys will show higher initial MV levels and smaller decreases in MV across adolescence, compared to adolescent girls.

Hypothesis 3 (H3): Youth raised in lower income families will show higher initial MV levels and smaller decreases in MV across adolescence, compared to youth raised in higher income families.

Parent Characteristics

Past research has focused on peer socialization of youth's materialism (e.g., Banerjee & Dittmar, 2008; Shi et al., 2016) as peers become the primary socialization agent in this stage (e.g., Moschis & Churchill, 1978; Shi & Xie, 2012). Note that parents are still powerful socializing agents given their cumulative influences over time (Pinquart, 2017). This is particularly true for Mexican-origin adolescents considering that a central Mexican traditional value (i.e., familism) highlights the importance of family responsibility, family cohesion, and respect toward parents (Perez-Brena et al., 2012; Sabogal et al., 1987). In this study, we examine the roles of parent MV, familism, and education in predicting Mexican-origin youth's MV trajectory. Both maternal and paternal influences on adolescent development will be examined, which is important because paternal influences are often neglected in developmental research (e.g., Padilla-Walker et al., 2016; Telzer, 2011).

Given the intergenerational transmission of values and a shared, culturally structured home environment (Knight et al., 2011), individuals from the same family are likely to have overlapping values such as materialism. Consistent with this idea, several studies have found positive associations between parent and child materialism (e.g., Ahuvia & Wong, 2002; Chaplin & John, 2010; Goldberg et al., 2003). However, we know of no longitudinal research examining whether parent materialism is associated with developmental changes in adolescent materialism.

[1] A relatively stronger emphasis on material possessions, such as appearance and image-relevant purchases, has been reported in female youth's self-identity and sense of self-worth as compared to male youth's (e.g., Dittmar, 2008; Usmiani & Daniluk, 1997). However, it is not clear whether female youth are more likely to endorse materialistic values than males, or female youth endorse values of beauty (e.g., physical attractiveness and body image) more than male youth.

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Moreover, social support from parents is influential on youth's identity development (e.g., Granott et al., 2002; Pinquart, 2017). As parents with high levels of familism value the importance of family as a source of social support, social connection, and social reference (e.g., Padilla-Walker et al., 2020), we examine parent familism as an antecedent of the development of adolescents' MV. Research indicates that youth growing up in more emotionally supportive families with higher parental warmth tend to report lower levels of materialism (e.g., Goldberg et al., 2003; Rindfleisch et al., 1997; Roberts et al., 2005). Through more supportive parenting, children develop a stronger sense of self and are lower in materialism (Chaplin & John, 2010). Conversely, when college students are asked to think of a contingently accepting person who would reject certain features of their behaviors and personalities, they report an increased extrinsic aspiration, which is a sign of materialism (Sheldon & Kasser, 2008). Therefore, it is possible that youth raised by parents with higher familism would be associated with lower levels of endorsement of MV.

In addition, parent education level may affect the development of youth's MV. When parents have a higher education level, children tend to endorse MV less (Chang & Arkin, 2002), perhaps because educated parents' communications with children about consumption, purchases, and advertising provide children with more nurturing during the consumer socialization process. Findings consistently show that mothers with higher levels of education engage more in the scaffolding interactions are more responsive and less punitive during the parent–child interaction (e.g., Carr & Pike, 2012; Neitzel & Stright, 2004). When parents adopt a more nurturing communication style and involve children more in discussions about commercials and purchases, children's MV tend to be less affected by exposure to external advertising (Buijzen & Valkenburg, 2005). Therefore, it can be inferred that youth raised by more educated parents are likely to show lower levels of endorsement of MV than youth with less educated parents.

Given that parents' MV, familism, and education level should capture an enduring child-rearing philosophy and a daily interactional style with their children, it is reasonable to assume that these characteristics may be associated with developmental *changes* in youth's MV over time. Based on the preceding discussion, we hypothesize that youth whose parents exhibit higher levels of MV may experience less of a decline in MV over time due to the cumulative transmission of values across generations. Furthermore, adolescents raised by parents with higher levels of familism and/or education may exhibit lower levels of MV at the outset of adolescence and may further demonstrate smaller developmental changes in MV across adolescence due to increased parental support, which may serve to reduce their openness to external peer influences. Thus, we predict that:

Hypothesis 4 (H4): Youth raised by parents with higher MV levels will show higher initial levels and smaller decreases in MV across adolescence, compared to youth raised by parents with lower MV levels.

Hypothesis 5 (H5): Youth raised by parents with higher familism levels will show lower initial levels and smaller decreases in MV across adolescence, compared to youth raised by parents with lower familism levels.

Hypothesis 6 (H6): Youth raised by more educated parents will show lower initial levels and smaller decreases in MV across adolescence, compared to youth raised by less educated parents.

Prospective Associations Between the Developmental Trajectory of MV Across Adolescence and Life Satisfaction in Early Adulthood

MV are fundamental beliefs about the importance and values of material acquisition and possessions in daily life (Richins & Dawson, 1992). The development of MV during adolescence is posited as a guiding force for future buying, spending, and other behaviors and decisions relevant to material acquisition and possessions. Consequently, MV in adolescence should play a significant role in young adult life satisfaction, especially during the transition from adolescence to early adulthood. Life satisfaction, an important component of subjective well-being (Diener et al., 1985), has been defined as a positive evaluation of the conditions of one's life. In this study, we focus on understanding how the development of Mexican-origin adolescent MV will longitudinally associate with their general life satisfaction in early adulthood. Given that life satisfaction changes dynamically over time (Willroth et al., 2021), we will address this issue with the investigation of the trajectory of life satisfaction from late adolescence (age 17) to young adulthood (age 23).

Two forms of goal pursuit—extrinsic and intrinsic—that differentially influence life satisfaction have been proposed in self-determination theory (SDT; R. M. Ryan & Deci, 2000). Several correlational and experimental studies suggest that intrinsic goal pursuit (e.g., pursuit of close relationships, caring for one's community, physical health) benefits subjective well-being, whereas extrinsic goal pursuit (e.g., pursuit of social recognition, wealth, beauty) does not (e.g., Bradshaw et al., 2023; Kasser & Ahuvia, 2002), even when there is a match between value orientation and values emphasized in the environment (Vansteenkiste et al., 2006).

Materialism is conceptualized as a specific form of extrinsic goal pursuit that involves the unbridled acquisition of material goods (Heaney et al., 2005). Materialistic people typically invest fewer resources in activities that are related to intrinsic goal pursuit, which has been shown to be more beneficial to one's subjective well-being and better satisfy the basic psychological needs of competence, autonomy, and relatedness than the pursuit of extrinsic goals (e.g., R. M. Ryan & Deci, 2000; Sheldon & Schüler, 2011). Moreover, materialism is related to detrimental and risky consumer behaviors, such as compulsive and impulsive buying (e.g., Islam et al., 2017; Ruvio et al., 2014; Yoon & Kim, 2016), overspending and becoming consumer indebtedness (e.g., de Matos et al., 2019; Watson, 2003). As proposed in SDT (R. M. Ryan & Deci, 2000), higher materialism should be associated with lower levels of subjective well-being.

An extensive body of research has found that a desire for wealth, possessions, and consumption is negatively related to personal well-being in a wide range of societies with diverse cultural and/or political contexts (e.g., Dittmar et al., 2014; Ku, 2015). For example, Ku (2015) found that materialism negatively affects life satisfaction and vice versa over time for older adolescents. L. Ryan and Dziurawiec (2001), in a cross-sectional study of Australian adults, found that individuals who are higher in materialism are

less satisfied with their "life as a whole" and with specific "life domains" than those who are lower in materialism. A negative association of materialism with general life satisfaction is identified in a cross-sectional study of adolescents with age 14–18 from United Kingdom (Barnfield, 2020). Kasser (2005) found that higher materialism is associated with greater anxiety, lower happiness, and poorer self-esteem in a sample of 11–18 years old. A meta-analysis of the literature on materialism and well-being shows a small but consistent negative association ($r = -.15$, $-.19$ after correction for measurement error) between various measures of materialism and various aspects of well-being (Dittmar et al., 2014). A very recent meta-analysis indicates a negative association of wealth aspiration as relative centrality (i.e., the mean for wealth aspiration minus the mean across all aspirations) with well-being ($r = -.16$; Bradshaw et al., 2023). Research using qualitative and experimental methods has also found that less material consumption relates to higher well-being and life satisfaction (e.g., Lee & Ahn, 2016; Moldes & Ku, 2020). Although a few studies have found a positive relationship between materialism and life satisfaction (e.g., Ger & Belk, 1999; Kuanr et al., 2020), a vast majority of research has demonstrated that higher levels of MV are associated with lower levels of life satisfaction. Thus, we predict:

> *Hypothesis 7 (H7):* A higher initial level of MV in adolescence will be associated with a lower initial level of life satisfaction in early adulthood, compared to a lower initial level of MV in adolescence.

There is little relevant research and theory to guide hypotheses about how *changes* in MV across adolescence might be associated with initial levels and changes in life satisfaction across early adulthood. Kasser et al. (2014), using data from three different longitudinal studies, found that changes in materialism are negatively associated with changes in well-being when materialism and well-being are assessed longitudinally across the same developmental period. Specifically, when people increase in MV over time, their well-being tends to decrease across the same time period, resulting in lower initial levels of well-being for the subsequent developmental period. This pattern replicates across different time intervals (6 months, 2 years, 12 years) and samples (i.e., a racially and socioeconomically diverse sample of American adolescents transitioning into adulthood, American college graduates entering the workforce, and Icelandic adults in the context of the 2008 economic collapse of Iceland). Thus, we predict:

> *Hypothesis 8 (H8):* Youth who show larger decreases in MV across adolescence will have higher initial levels in life satisfaction in early adulthood, compared to youth who show smaller decreases in MV across adolescence.

Researchers have explored the underlying mechanism for the negative association of MV with life satisfaction. A substantial body of evidence indicates that adults with higher (vs. lower) MV exhibit more ambivalence in their sense of self-worth (Frost et al., 2007), scoring lower on self-concept certainty (Noguti & Bokeyar, 2014; Rindfleisch et al., 2009) and self-concept clarity (Reeves et al., 2012). Individuals who prioritize MV over other values tend to express more self-doubt (Chang & Arkin, 2002) and lower levels of self-esteem (Park & John, 2011; Ruvio et al., 2014), particularly intrinsic self-esteem related to personal growth and being true to oneself (Nagpaul & Pang, 2017). Interventions that enhance feelings of security cause decreases in materialism, as adolescents who experience temporary boosts in self-esteem become less materialistic soon after (Chaplin & John, 2007). Therefore, efforts to reduce materialism would be successful to the extent that they improve factors such as a sense of security and self-esteem, which are in turn contributive to promoting one's life satisfaction (e.g., Butkovic et al., 2020). Considering that factors such as a sense of security and self-esteem are suggested as relatively stable (Kuster & Orth, 2013) and posited as part of preexisting self-characteristics for the subsequent development (e.g., Fischer, 1980; Harter, 1999), we predict that:

> *Hypothesis 9 (H9):* Youth who show larger decreases in MV across adolescence will experience more increases (or smaller decreases) in life satisfaction in early adulthood, compared to youth who show smaller decreases in MV across adolescence.

Moderating Effects of Youth Familism on the Associations Between Adolescent MV and Early Adult Life Satisfaction

Familism, a central value in Mexican-origin families, develops simultaneously in adolescence. As SDT (R. M. Ryan & Deci, 2000) distinguishes intrinsic goals from extrinsic ones, familism belongs to strong intrinsic goals considering that it involves emphases such as the desirability to maintain close relationships (emotional support), the importance of tangible caregiving (obligation to family), and the reliance on communal interpersonal reflection to define the self (family as a referent; Knight et al., 2011). Based on the proposed improvement in subjective well-being induced by the pursuit of intrinsic goals (R. M. Ryan & Deci, 2000), familism should be a protective factor in attenuating the negative association of MV with Mexican-origin youth's life satisfaction.

Empirically, youth's familism values are associated with positive family relationship dynamics and subjective well-being in adolescence (e.g., Fuligni et al., 1999; Updegraff et al., 2005). Research has documented the importance of familism for reducing mental health problems and risk behaviors (e.g., G. L. Stein et al., 2014; Wheeler et al., 2017). In specifics, youth's strong familism values are linked with parent–youth high-quality relationship (Padilla-Walker et al., 2020). As adolescent life satisfaction is positively associated with perceived parental support (Burke & Weir, 1979; Young et al., 1995), cohesive family relationships (Manzi et al., 2006), and perceived quality of attachment to parents (Greenberg et al., 1983), we predict that:

> *Hypothesis 10 (H10):* Familism will moderate the associations of MV with life satisfaction, such that the hypothesized associations between MV and life satisfaction specified in H7–H9 should be less strong or nonexistent for youth with higher (vs. lower) levels of familism.

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Moderating Effects of Gender, Nativity Status, and Household Income

The hypothesized associations of adolescent MV with Mexican-origin youth's life satisfaction should vary with youth characteristics. Based on social roles in which boys are proposed as agent-oriented, whereas girls are communal-oriented (e.g., J. A. Stein et al., 1992), it is argued that it is less acceptable for girls to be materialistic (e.g., Beutel & Marini, 1995). Dittmar et al. (2014) found that the link between materialism and lower well-being is stronger when women make up a greater proportion of the sample. With the recruitment of Mexican American women and men residing in a major Southwestern city (Trim, 2008), materialism was found to significantly predict depressive symptoms only among female participants. Consequently, we hypothesize that:

> *Hypothesis 11 (H12):* Hypothesized associations between MV and life satisfaction specified in H7–H9 will be stronger for girls than for boys.

There is little theory or research that could be used to generate hypotheses regarding nativity status. Consequently, nativity as a moderator will be examined without any hypotheses.

Based on aspiration theory (e.g., Carp & Carp, 1982), which asserts that the gap between what they actually have and what people aspire to would affect their life satisfaction, bad current conditions (e.g., low household income) coupled with high aspirations would negatively influence their perceived life satisfaction. Consistently, household income moderates the negative impact of how much importance young adults attach to wealth on overall life satisfaction (Nickerson et al., 2003): In lower income households, participants with a higher financial goal have lower overall life satisfaction than those with a lower financial goal, whereas in higher income households, there is little difference in overall life satisfaction between participants with higher versus lower financial goals. Ruvio et al. (2014) have demonstrated a stronger relation between life satisfaction and MV in high (vs. low) economic hardship. Thus, we predict:

> *Hypothesis 12 (H12):* Hypothesized associations between MV and life satisfaction specified in H7–H9 will be stronger for youth growing up in families with lower (vs. higher) income levels.

In summary, the present study uses data from a 14-year longitudinal study of Mexican-origin youth to address three broad aims. Aim 1 focuses on documenting the normative trajectory of MV across adolescence (ages 10–16; H1) and the codevelopment of MV and familism. Aim 2 focuses on identifying antecedent factors that predict the normative trajectory of MV across adolescence, including youth gender (H2), nativity, household income (H3), parent MV (H4), familism (H5), and education level (H6). Finally, Aim 3 investigates the associations of adolescent MV with life satisfaction in early adulthood (H7–H9) and the potential moderating roles of youth familism (H10), gender (H11), nativity, and household income (H12). The present study will extend prior research by furthering our understanding of how MV develop across adolescence, identifying antecedent factors associated with individual differences in the MV development, and documenting the prospective associations of adolescent MV with life satisfaction in early adulthood.

Method

Participants and Procedure

The present study used data from the California Families Project, an ongoing longitudinal study of Mexican-origin youth and their parents ($N = 674$).[2] Children were drawn at random from rosters of students from the Sacramento and Woodland, California school districts, in 2006–2007. The focal child had to be in the fifth grade, of Mexican origin, and living with their biological mother in order to participate in the study. Approximately 72.6% of the eligible families agreed to participate in the study. The children (50% female) were assessed annually from age 10 to 19 and then at ages 21 and 23. The present study used data when the children were ages 10 ($M_{age} = 10.86$, $SD = 0.50$), 12 ($M_{age} = 12.81$, $SD = 0.49$), 14 ($M_{age} = 14.75$, $SD = 0.49$), 16 ($M_{age} = 16.80$, $SD = 0.51$), 17 ($M_{age} = 17.73$, $SD = 0.52$), 18 ($M_{age} = 18.75$, $SD = 0.51$), 19 ($M_{age} = 19.86$, $SD = 0.52$), 21 ($M_{age} = 21.74$, $SD = 0.73$), and 23 ($M_{age} = 23.18$, $SD = 0.58$). Retention rates compared to the original sample are as follows: 86% (age 12), 90% (age 14), 89% (age 16), 89% (age 17), 87% (age 18), 87% (age 19), 80% (age 21), and 81% (age 23).

Participants were interviewed in their homes in Spanish or English, depending on their preference. Interviewers were all bilingual and most were of Mexican heritage. Sixty-three percent of mothers and 65% of fathers had less than a high school education (*Mdn* = ninth grade for both mothers and fathers); median total household income was between \$30,000 and \$35,000 (overall range of income ≤ \$5,000 to >\$95,000). Regarding nativity status, 83.6% of mothers and 89.4% of fathers were first generation, and 16.4% of mothers and 10.6% of fathers were either second or third generation. At Wave 1, 549 of the families were two-parent households and 124 families were single parent.

Ethics Statement

California Families Project was approved by the University of California, Davis Institutional Review Board (Protocol Number 217484-21).

Measures

Materialistic Values

The MACVS (Knight et al., 2011) was developed through focus groups of immigrant and U.S.-born Mexican-origin adolescents and adults who identified values that they ascribed to Mexican and American culture. Children completed the MACVS at ages 10, 12, 14, and 16. The Materialistic Values Scale includes the following five items: "Money is the key to happiness"; "Owning a lot of nice things makes one very happy"; "The more money one has, the more

[2] This project involves secondary data analysis of the California Families Project (CFP). No previous published work has used CFP data to examine the development of materialistic values or its association with life satisfaction. For a complete list of CFP publications, see https://www.californiafamilieproject.org/publications.html. The sample size was determined prior to the CFP data collection.

respect they should get from others"; "The best way for a person to feel good about himself/herself is to have a lot of money"; and "Children should be taught that it is important to have a lot of money." Ratings were made on a 4-point scale from 1 (*not at all*) to 4 (*very much*). We computed a latent variable of MV at each wave with the five items as indicators.

Parents completed the MACVS when children were ages 10, 12, and 14, but the present study focused on the age 10 data. We computed a latent variable of MV at age 10 using the five items as indicators, separately for moms and dads. Reliabilities and descriptive statistics for child, mom, and dad Materialistic Values Scales are shown in Table 1.

Familism

The MACVS includes three subscales related to familism. The Family Support subscale (six items) measures one's desirability to have a close connection with the family members. An example item is "Family provides a sense of security because they will always be there for you." The Family Obligation subscale (five items) measures the extent to which one values tangible caregiving in the family. An example item is "Children should be taught that it is their duty to care for their parents when their parents get old." The Family Referents subscale (five items) measures the tendency of defining one's self-identity by communal interpersonal reflection. An example item is "When it comes to important decisions, the family should ask for advice from close relatives." For child (at ages 10, 12, 14, and 16), mother (at child's age 10), and father (at child's age 10), we computed a latent factor of overall familism using the three subscales as indicators. Reliabilities and descriptive statistics for child, mom, and dad familism scales are shown in Table 1.

Life Satisfaction

To assess life satisfaction, we used a single-item measure of global life satisfaction (i.e., "How satisfied or dissatisfied are you with your life as a whole?"; Campbell et al., 1976). Youth rated this item at ages 17, 18, 19, 21, and 23, using a 5-point Likert scale ranging from 1 (*completely dissatisfied*) to 7 (*completely satisfied*). Means and standard deviations are $M = 4.22$, $SD = 0.71$ (age 17); $M = 4.16$, $SD = 0.72$ (age 18); $M = 4.08$, $SD = 0.74$ (age 19); $M = 4.01$, $SD = 0.75$ (age 21); $M = 4.02$, $SD = 0.70$ (age 23).

Demographic Variables

Household Income. At Wave 1, the total annual household income was reported by the mothers using a 20-point ordinal response scale, with response options increasing in $5,000 increments (1 = "less than $5,000," 2 = "$5,000–$10,000, …, up to 20 = "95,000 or more"). The median total household income was $32,500.

Nativity Status. We created a dichotomous nativity status variable comparing youth born in Mexico (29%) versus the United States (71%).

Parent Education Level. Mothers and fathers each reported their total years of education at Wave 1. The median education level for both mothers and fathers was ninth grade.

Data Analysis Plan

All data cleaning and analyses were conducted in R via RStudio Version 1.4.1717 using full information maximum likelihood estimation to account for missing data (Allison, 2003; Schafer & Graham, 2002). We used the following R packages to analyze the data and visualize the results: *psych* (Revelle, 2020), *lavaan* (Rosseel, 2012), *ggplot2* (Wickham, 2016), *semTools* (Jorgensen et al., 2022), and *lm.beta* (Behrendt, 2014). Model fit was assessed via comparative fit index (CFI) and root-mean-square error of approximation (RMSEA). We interpreted good fit as values greater than or equal to .95 for CFI and less than or equal to .06 for RMSEA (Hu & Bentler, 1999). We compared models based on a holistic evaluation of multiple criteria: difference in CFI above the threshold of .01, difference in RMSEA was above the threshold of .015, and chi-square difference test (Chen, 2007; Cheung & Rensvold, 2002; Meade et al., 2006).

Longitudinal Measurement Invariance Analyses

We conducted tests of longitudinal measurement invariance for all latent variables measured across waves. We compared three measurement models: (a) configural model, which freely estimated the factor loadings of items/parcels at each assessment wave; (b) weak invariance model, which constrained the factor loadings of corresponding items/parcels to be time invariant; (3) strong invariance model, which constrained the factor loadings and intercepts to be time invariant. If the fit of more constrained models was better than or similar to that of the lesser constrained models, we concluded that the factor structure of the latent construct was the same over time.

Table 1
Descriptive Statistics and Reliability Across Measurement Items for Materialistic Values and Familism

	Age 10		Age 12		Age 14		Age 16	
Variable	*M* (*SD*)	α/ω	*M* (*SD*)	α/ω	*M* (*SD*)	α/ω	*M* (*SD*)	α/ω
Materialistic values								
Child	2.03 (0.71)	.77/.79	1.84 (0.63)	.79/.82	1.76 (0.60)	.79/.81	1.80 (0.59)	.79/.81
Mom	1.93 (0.62)	.76/.80						
Dad	2.16 (0.67)	.79/.82						
Familism								
Child	3.63 (0.31)	.81/.83	3.60 (0.33)	.86/.88	3.52 (0.36)	.87/.88	3.44 (0.39)	.88/.89
Mom	3.55 (0.31)	.82/.84						
Dad	3.55 (0.30)	.81/.83						

Note. α = alpha reliability; ω = omega total reliability.

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When the strong invariance model fit significantly worse than the weak invariance model, we compared a "partial" strong invariance model with the weak invariance model, given that it is best to retain a strong invariant model whenever possible (Widaman et al., 2010). Results of longitudinal measurement invariance analyses were shown in Supplemental Table S1. For subsequent latent growth curve (LGC) analyses, we retained the optimal measurement invariance on the latent variables in the growth curve models.

Hypotheses Test for Aim 1: Normative Trajectory of Child MV Across Adolescence

To test H1, we ran univariate LGC models with four time points (at children ages 10, 12, 14, and 16) to examine mean-level change in child MV. We conducted a series of LGC model comparisons and evaluated parsimony and changes in fit indices to identify the optimal growth trajectory. We compared three models: (a) no growth model, which estimated only the intercept; (b) linear growth model, where the slope factor loadings were fixed to be 0, 0.3333, 0.6666, and 1 at age 10, 12, 14, and 16; and (c) latent basis model, where the slope factor loadings at ages 10 and 16 were fixed to be 0 and 1, and the loadings of the middle waves were freely estimated. If the model did not fit significantly worse, we chose the more parsimonious model. The model comparison procedure helps us to select the model that describes the normative trajectory of MV. Results of LGC model comparisons were shown in Supplemental Table S2.

Hypotheses Test for Aim 2: Antecedent Factors That Predict the Normative Trajectory of Child MV Across Adolescence

We tested H2–H6 with LGC models where the latent intercept and slope of child MV were regressed on each of the antecedent factors (youth gender, household income, parent MV, parent familism, and parent education level when the child was age 10). These antecedent factors were examined in separate models. In each LGC model, the covariance between the level and the slope was allowed so that the effect of the antecedent factor on the slope was not confounded by the level.

Hypotheses Test for Aim 3: The Impact of Child MV on Life Satisfaction and the Moderating Effects of Child Familism, Gender, Nativity, and Household Income

First, we followed the same procedure used in H1 to select the best univariate LGC model of child life satisfaction from age 17 to 23. The only difference is that for the linear growth model of life satisfaction, the slope factor loadings were fixed to be 0, 1/6, 1/3, 2/3, and 1 to account for the unequal time intervals. Model statistics for familism and life satisfaction were shown in Supplemental Table S3. H7–H9 were tested by running a bivariate LGC model where the intercept and slope of child life satisfaction were regressed on the intercept and slope of child MV. The intercept and slope of the same variable were allowed to covary.

H10 was tested by running moderated multiple regression analyses using factor scores saved from the optimal LGC models. First, we saved the intercept and slope factor scores produced by the LGC models of child MV and child familism. Second, we computed all possible interactions between the intercepts and slopes: Intercept of MV × Intercept of Familism, Intercept of MV × Slope of Familism, Slope of MV × Intercept of Familism, and Slope of MV × Slope of Familism. Third, we conducted a series of moderated multiple regression analyses in which the level of the life satisfaction trajectory was predicted by each interaction term, along with the two corresponding main effects, and a separate series of analyses in which the slope of the life satisfaction trajectory was predicted by the same set of interaction terms and main effects.

H11 was tested by running multigroup structural equation modeling analyses. We tested gender differences in three effects: the intercept of child MV predicting the intercept of life satisfaction, the slope of child MV predicting the intercept of life satisfaction, and the slope of child MV predicting the slope of life satisfaction. When testing one of the three effects, the other two effects were fixed to be the same for both genders. H12 was tested by the same procedure used to test H10. Specifically, we computed the interaction between household income and the factor scores for the intercept and slope of MV, and then examined each interaction term, and the corresponding main effects, as predictors of the level and slope of life satisfaction.

Results

Table 2 shows intercorrelations among all variables. Two-year rank-order stabilities of MV ranged from .54 to .61, indicating moderate to high levels of rank-order stability. The 6-year rank-order stability of MV from age 10 to 16 was only .25, indicating somewhat weak stability of rank across adolescence. We corrected α for multiple tests and interpreted $p \leq .01$ as significant.

Aim 1: Normative Developmental Trajectory of MV Across Adolescence

Child MV demonstrated partial strong invariance (Supplemental Table S1). Trajectories of child MV are shown in Figure 1 (see Table 3 for model parameters). We retained the linear growth model of child MV for parsimony.[3] The linear model showed acceptable model fit (CFI = .94, RMSEA = .044, 90% CI [.039, .050]; Table 4). H1 was supported. Child MV decreased from age 10 to 16 ($\beta = -.71$, $p < .001$), indicating that youth in this study become less materialistic over the course of adolescence.

We conducted analyses of the codevelopment of child MV and familism. The univariate LGC model of child familism from age 10 to 16 was identified following the similar procedure for testing H1. The strong invariant model (Supplemental Table S1) was retained for child familism. The linear growth model was identified as the optimal model (Supplemental Table S2; see Supplemental Table S3 for model statistics). Child familism decreased from age 10 to 16 ($\beta = -0.92$, $p < .001$).

In the first bivariate LGC model (as presented in Supplemental Table S4), the slope factor loadings of MV and familism were specified based on the selected univariate LGC models. The intercepts and slopes of MV and familism were allowed to covary

[3] Model comparison showed that the linear growth model, the quadratic model, and the latent basis model of child materialistic values were not significantly different regarding model fit (ΔCFI < .010, ΔRMSEA < .015), but all three models fit significantly better than the no-growth model (see Supplemental Table S2). Therefore, we retained the linear growth model as showing the normative trajectory of MV.

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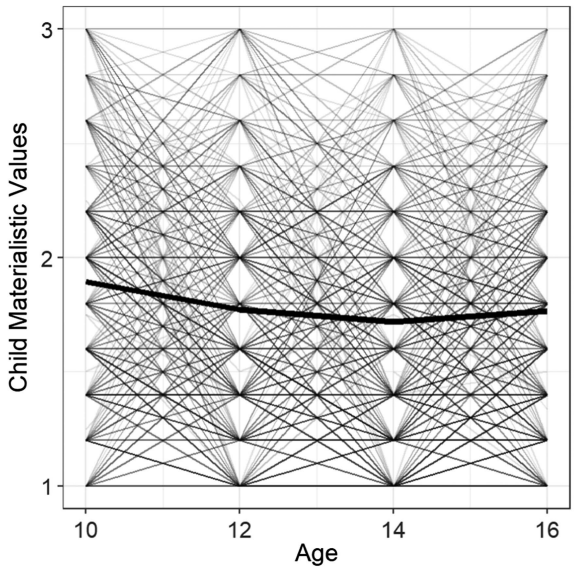
Table 2
Intercorrelations Among Study Variables

Variable	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22
1. Child materialistic values (age 10)	—																					
2. Child materialistic values (age 12)	.65*	—																				
3. Child materialistic values (age 14)	.50*	.72*	—																			
4. Child materialistic values (age 16)	.32*	.55*	.65*	—																		
5. Mom materialistic values (age 10)	.19*	.19*	.18*	.10*	—																	
6. Dad materialistic values (age 10)	.11*	.18*	.16*	.16*	.15*	—																
7. Child familism (age 10)	.04	−.08*	−.10*	−.06	.00	.01	—															
8. Child familism (age 12)	.04	.02	−.06	−.07	−.01	.03	.49*	—														
9. Child familism (age 14)	.06	.03	−.08*	−.06	.03	.03	.40*	.67*	—													
10. Child familism (age 16)	.08*	−.00	−.04	−.04	.05	.06	.40*	.61*	.62*	—												
11. Mom familism (age 10)	.08	.06	.11*	.05	.27*	.09	.10*	.11*	.07	.15*	—											
12. Dad familism (age 10)	.07	.07	.08	.01	.07	.28*	.12*	.13*	.08	.11*	.12*	—										
13. Child life satisfaction (age 17)	.06	.01	−.05	−.11*	.05	−.01	.05	.13*	.15*	.22*	.04	−.03	—									
14. Child life satisfaction (age 18)	.12*	.06	.03	−.01	.01	−.01	.05	.12*	.11*	.13*	.06	.04	.43*	—								
15. Child life satisfaction (age 19)	.00	−.06	−.08	−.11*	−.00	−.05	.06	.10*	.10*	.17*	−.01	.04	.40*	.47*	—							
16. Child life satisfaction (age 21)	.05	.09*	.06	−.00	−.02	.06	−.00	.09*	.12*	.13*	.06	.13*	.27*	.31*	.36*	—						
17. Child life satisfaction (age 23)	.07	.07	.01	.02	−.00	−.05	.05	.07	.14*	.18*	.02	.04	.24*	.35*	.35*	.38*	—					
18. Mom education	−.16*	−.14*	−.10*	−.03	−.32*	−.23*	−.02	−.01	.00	−.02	−.14*	−.10*	−.02	−.04	−.01	−.02	−.03	—				
19. Dad education	−.13*	−.13*	−.15*	−.11*	−.14*	−.36*	.02	−.00	−.02	−.07	−.10*	−.22*	−.03	−.08	−.03	−.03	−.03	.39*	—			
20. Household income	−.09*	−.12*	−.08*	−.00	−.14*	−.20*	.03	.07	.05	.01	−.08*	−.14*	.01	−.00	.04	−.02	−.03	.31*	.24*	—		
21. Gender	.09*	.16*	.19*	.20*	−.00	−.04	−.02	.01	.03	−.03	−.01	−.02	.02	−.03	.01	.08*	.02	−.03	.01	.03	—	
22. Nativity	.03	.05	.02	.01	−.03	−.05	−.04	−.05	−.02	−.04	−.04	−.07	−.07	−.05	−.08*	−.02	−.07	.17*	.10*	.24*	−.07	—

Note. For child gender, 0 = female, 1 = male. For nativity status, 0 = born in Mexico, 1 = born in the United States.
* $p < .05$.

Figure 1
Normative Growth Trajectories for Child Materialistic Values



Note. The *y*-axis ranges from 1 to 3 given that very few trajectories were above 3.

with each other. Results showed that initial levels of MV and familism were not associated (β = 0.04, p = .63), and neither were changes in MV and familism (β = −0.12, p = .36). MV levels at age 10 were not associated with changes in familism (β = .134, p = .202). Higher levels of familism at age 10 predicted slower growth of MV from age 10 to 16 (β = −.24, p = .025) but the effect was not significant after correction for multiple comparisons.

Table 3
Model Statistics From the Univariate Latent Growth Curve Model for Child Materialistic Values

Univariate latent growth model	Materialistic values (child)
Slope	
β_1	0
β_2	1
β_3	2
β_4	3
Means	
Intercept	0
Slope	−.084***
Variances	
Intercept	.138***
Slope	.014***
$\text{Covariance}_{\text{Intercept, Slope}}$	−.027***
Goodness-of-fit	
χ^2 (*df*)	398.947 (172)
RMSEA [90% CI]	.044 [.039, .050]
CFI	.940

Note. Values are unstandardized coefficients for the models. χ^2 = chi-square; *df* = degrees of freedom; RMSEA = root-mean-square-error of approximation; 90% CI = 90% confidence interval; CFI = comparative fit index. This model was a second-order latent growth curve model. For the model to converge, the intercept mean had to be set to 0.
*** p < .001.

Aim 2: Antecedent Factors of the Trajectory of Child MV Across Adolescence

The results of LGC analyses are presented in Table 4. H2 was not supported. Boys and girls did not differ in either the level or the slope of their MV trajectory (ps > .016), although a weak effect indicated boys were higher in initial MV (measured at age 10 in the present study) than girls was detected (β = .17, p = .017), the effect was not significant after correction for multiple comparisons. U.S.-born children did not differ from Mexico-born children in their level of MV at age 10 or in the slope of their MV trajectory across adolescence (ps > .30). H3 was partially supported. Higher household income was associated with lower child MV at age 10 (β = −.32, p = .001). In addition, higher household income at a child's age of 10 was associated with a smaller decrease in MV from age 10 to 16 (β = .28, p = .03) but the effect was not significant after correction for multiple comparisons.

H4 was partially supported. Higher mom and dad MV were concurrently associated with higher child MV at age 10 (β = .28, p < .001; β = .22, p = .003, respectively). Parent MV was not associated with the slope of the child MV trajectory across adolescence (ps > .02). Higher mom MV at the child's age 10 predicted smaller increases in child MV (β = −.18, p = .02), the effect was not significant after correction for multiple comparisons. H5 was not supported. Mom and dad familism were not associated with the level or slope of child MV (ps > .28). Last, H6 was partially supported. Children raised by more educated moms and dads tended to have lower MV at age 10 (β = −.56 for mom education and β = −.43 for dad education, p < .001). In addition, a more educated mom (but not dad) was associated with smaller decreases in child MV from age 10 to 16 (β = .54, p = .001).

Aim 3: Prospective Associations Between Adolescent MV and Early Adult Life Satisfaction, and the Moderating Effects of Child Familism, Gender, Nativity, and Household Income

The bivariate LGC model with MV and life satisfaction demonstrated acceptable model fit, χ^2 = 521.96, *df* = 288, CFI = .95, RMSEA = .035, 95% CI [.030, .040]. H7 was not supported because the association between child MV at age 10 and life satisfaction at age 17 was not significant (p = .33; Table 5). H8 was supported. A greater decrease in child MV from age 10 to 16 predicted a higher level of child life satisfaction at age 17 (β = −.36, p = .002). Inconsistent with H9, a more rapid decrease in child MV from age 10 to 16 was associated with a more rapid decrease in child life satisfaction from age 17 to 23 (β = .31, p = .036); the effect was not significant after correction for multiple comparisons.

Moderating Effects

When predicting the trajectory of child life satisfaction, the interaction between the factor scores of child MV and familism was not significant (ps > .04; Supplemental Table S5). H10 was not supported, except that an interaction effect between the slope of MV and the slope of familism was found on the slope of life satisfaction (β = .27, p = .038). The interaction effect was not significant after correction for multiple comparisons. In addition, the association between the MV and life satisfaction trajectories did not differ

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Table 4
Predictors of the Child Materialistic Values Trajectory

Child materialistic values trajectory	Antecedent factor	*B* (*SE*)	β	Wald's *z*	*p*
Model 1					
Level	Child gender	.10 (.04)	.17	2.38	.017
Slope	Child gender	.03 (.02)	.17	1.86	.063
Model 2					
Level	Child nativity	.04 (.04)	.10	1.03	.303
Slope	Child nativity	−.01 (.02)	−.08	−.65	.515
Model 3					
Level	Household income	−.01 (.00)	−.32	−3.30	.001
Slope	Household income	.00 (.00)	.28	2.19	.029
Model 4					
Level	Mom education level	−.03 (.01)	−.56	−4.50	<.001
Slope	Mom education level	.01 (.00)	.54	3.35	.001
Model 5					
Level	Dad education level	−.02 (.00)	−.43	−4.22	<.001
Slope	Dad education level	.00 (.00)	.24	1.72	.086
Model 6					
Level	Mom materialistic values (age 10)	.17 (.04)	.28	4.62	<.001
Slope	Mom materialistic values (age 10)	−.03 (.02)	−.18	−2.29	.022
Model 7					
Level	Dad materialistic values (age 10)	.13 (.04)	.22	2.94	.003
Slope	Dad materialistic values (age 10)	−.00 (.02)	−.01	−.07	.943
Model 8					
Level	Mom familism (age 10)	.05 (.10)	.03	.57	.571
Slope	Mom familism at (age 10)	.01 (.04)	.03	.34	.732
Model 9					
Level	Dad familism (age 10)	.12 (.11)	.08	1.07	.284
Slope	Dad familism (age 10)	−.02 (.04)	−.05	−.52	.601

Note. $p \leq .01$ is deemed as significant. *SE* = standard error.

significantly for boys versus girls or for youth born in the United States versus Mexico (Supplemental Table S6). H11 was not supported. Finally, different from H12, we did not find evidence for a significant moderation effect of household income (ps > .14; Supplemental Table S5).

General Discussion

Following the framework of social identity development with the adoption of a neo-Piagetian approach, this study examined the trajectory of MV across adolescence (ages 10–16) and its prospective associations with life satisfaction in early adulthood (ages 17–23). Data from a 14-year longitudinal study of 674 Mexican-origin youth were analyzed. On average, MV decreased significantly from early to late adolescence. Youth raised in families with higher household income, higher maternal and paternal education, and lower maternal and paternal MV tended to be less materialistic. In addition, higher maternal education was associated with *smaller* decreases in MV from age 10 to 16.

Furthermore, youth who increased in MV from age 10 to 16 reported lower levels of life satisfaction at age 17, but no strong associations were found between the slope of MV across adolescence and the slope of life satisfaction in early adulthood. The proposed moderators (familism, gender, nativity status, household income) failed to significantly influence the associations between MV and life satisfaction trajectories. We did, however, find a main effect of familism; youth who exhibited greater increases in familism from age 10 to 16 reported higher levels of life satisfaction at age 17 (β = .36, $p < .001$; Supplemental Table S4). Below, we discuss the findings in greater detail.

The Normative Trajectory of MV Across Adolescence

Transitions and important life changes have been recognized as particularly sensitive periods for social identity development (Harter, 1999). During the transition to adolescence, a critical developmental task is to establish a social identity among peers (Erikson, 1968). This task is particularly salient in early adolescence as the transition from childhood to adolescence is marked by a shift of the primary socialization agent from parents toward peers. With accumulated experiences of interacting with peers and improved competence in interpersonal skills, youth's reliance on external tangible resources, such as material possessions, to establish and maintain an identity among peers may decrease over time. Our findings support this proposition. Consistent with H1, MV decreased over time from early to late adolescence. This pattern is consistent with cross-sectional findings (Chaplin & John, 2007) and the proposed "temporary change toward immaturity" in early adolescence (Van den Akker et al., 2014, p. 737).

Moreover, as the interdependencies among identity developments in different domains are proposed as one major component of social identity development (Grotevant, 1987), we examined the codevelopment of MV and familism across adolescence. Findings indicated that in general, the initial levels and the slopes of MV

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Table 5
Model Statistics for the Bivariate LGC Model of Child Materialistic Values and Life Satisfaction

Bivariate LGC model	*b* (*SE*)	β	Wald's *z*	*p*
Means				
Child materialistic values intercept	0			
Child materialistic values slope	−.08 (.01)	−.68	−8.46	<.001
Child life satisfaction intercept	4.07 (.05)	8.20	85.41	<.001
Child life satisfaction slope	−.02 (.01)	−.22	−1.74	.083
Variances				
Child materialistic values intercept	.14 (.02)	1	7.34	<.001
Child materialistic values slope	.02 (.00)	1	4.77	<.001
Child life satisfaction intercept	.22 (.03)	.90	8.91	<.001
Child life satisfaction slope	.01 (.00)	.94	4.00	<.001
Covariances				
Child materialistic values intercept–child materialistic values slope	−.03 (.01)	−.62	−4.58	<.001
Child life satisfaction intercept–child life satisfaction slope	−.02 (.01)	−.43	−3.33	.001
Regression				
Child life satisfaction intercept				
Child materialistic values intercept	−.10 (.11)	−.08	−.97	.332
Child materialistic values slope	−1.49 (.47)	−.36	−3.13	.002
Child life satisfaction slope				
Child materialistic values intercept	.04 (.02)	.20	1.83	.068
Child materialistic values slope	.20 (.10)	.31	2.10	.036

Note. In this model, we fixed the intercept mean of child materialistic values to be 1. We fixed the intercept and slope variances of child materialistic values to be 1 s. The reason is that for child materialistic values, we fit the second-order latent growth curve models, whereas for child life satisfaction, we fit the first-order latent growth curves. $p \leq .01$ was deemed as significant. LGC = latent growth curve; *SE* = standard error.

and familism were not correlated, suggesting that Mexican-origin youth's MV and familism develop through different, independent processes across adolescence.

Antecedents of the MV Trajectory

Regarding individual differences in the MV trajectory, we did not find strong evidence for the effect of gender. Boys were more materialistic than girls ($\beta = .17$, $p = .017$; the effect was not significant after correction for multiple comparisons) and showed a tendency of slower decreases in MV than girls ($\beta = .17$, $p = .063$), which tends to follow the developmental pattern hypothesized in H2. In addition, nativity was not an influential factor. The developmental trajectory of MV for youth was similar across the first- and second+ generation immigrants.

Partially consistent with H3, higher household income was associated with lower levels of MV at the beginning of adolescence. This finding suggests that high household income protects Mexican-origin youth against the tendency of relying on material possessions as a way of dealing with the challenging developmental tasks in early adolescence (e.g., building their own social identity and making social comparisons with peers). However, different from H3, the protective effect of high household income appeared to fade over time, given that youth decreased in MV to a similar extent, regardless of whether they grew up in higher versus lower income households.

As the contexts of development are important in identity development (Grotevant, 1987), support from significant others such as parents should play a critical role in youth's development. Consistently, parent MV and education were found as significant antecedents of the trajectory of adolescent MV, which partially support H5 and H7. Higher levels of maternal and paternal education and lower levels of maternal and paternal MV were associated with lower MV among Mexican-origin adolescents at age 10. Moreover, higher levels of maternal education were associated with smaller decreases in MV across adolescence, whereas paternal education and maternal/paternal MV were not significantly associated with changes in MV across adolescence. The findings demonstrate that more positive family environments such as higher maternal/paternal education and lower maternal/paternal MV are associated with lower MV at the initial stage of adolescence when they just undergo a shift of the primary socialization agent from parents to peers (e.g., Moschis & Churchill, 1978; Shi & Xie, 2012).

Throughout adolescence, maternal education, but not paternal education, was associated with the development of Mexican-origin adolescent MV. These findings suggest that both mothers and fathers are important in influencing initial levels of adolescent MV, while over time, it is highly educated mothers who are more likely associated with the developmental changes in youth's MV. Note that higher maternal education was associated with smaller decreases in MV across adolescence. One interpretation is that well- (vs. poorly-) educated mothers often initiate and involve youth in nurturing and supportive communications and discussions. Youth raised by well- (vs. poorly-) educated mothers would thus be more mature in their endorsement of material possessions and less likely to change in their endorsement of MV across adolescence.

Different from H6, parent familism did not show significant effects on the development of Mexican-origin adolescent MV. With consideration of the abovementioned findings on the independent development of MV and familism among Mexican-origin adolescents, the nonsignificant effects of parent familism appear to provide indirect support for the pattern suggesting that MV and familism develop via different, independent processes across adolescence.

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The Prospective Associations of the Adolescent MV Trajectory With Life Satisfaction in Early Adulthood, and the Potential Moderating Effects of Familism and Youth Characteristics

It is worthwhile to note that investigation of MV is important in adolescence not just because MV represent a specific domain closely involved in social identity development among peers but also because MV are about individuals' fundamental beliefs in material possessions, having strong influences on their future consumption, purchases, and, furthermore, other important outcomes such as life satisfaction. Therefore, in this study, we examined the prospective associations of the MV trajectory across adolescence with life satisfaction in early adulthood. Based on SDT (R. M. Ryan & Deci, 2000) and an extensive body of research indicating that a desire for wealth, possessions, and consumption is negatively related to personal well-being (e.g., Dittmar et al., 2014; Ku, 2015), we proposed H7–H9.

The initial level of MV in adolescence (age 10) was not found to predict life satisfaction at age 17 or change in life satisfaction across early adulthood (age 17–23). This finding is different from H7, which was hypothesized based on the robust concurrent negative association between MV and life satisfaction observed in the literature (e.g., Dittmar et al., 2014; Ku, 2015). The nonsignificant association between MV's initial level in early adolescence and the trajectory of life satisfaction in early adulthood suggests that a youth's MV level at a relatively early time point, namely age 10, is almost irrelevant to life satisfaction around seven and more years later (i.e., at age 17–23).

Regarding *change* in adolescent MV, greater decreases in MV across adolescence (age 10–16) predicted higher life satisfaction in early adulthood (age 17). This finding is consistent with H8, which, based on the co-occurrence of decreased MV and improved life satisfaction identified among young individuals (e.g., Chaplin & John, 2007; Donnelly et al., 2016; Kasser et al., 2014), states that greater decreases in MV across adolescence would be associated with higher initial levels of life satisfaction in early adulthood. This finding contributes to a greater understanding of potential antecedents of life satisfaction in early adulthood, suggesting materialism as a modifiable risk factor for improving the life satisfaction of young adults.

However, we did not find the predicted (H9) association between more decreases in MV across adolescence (age 10–16) and larger increases in life satisfaction across early adulthood (age 17–23). Rather, we found a tendency opposite to the hypothesis: More decreases in adolescent MV were associated with more decreases in life satisfaction in early adulthood ($\beta = .31$, $p = .036$), whereas the effect was not significant after correction for multiple comparisons. We speculate that one alternative would be that, with decreased MV across adolescence, the relevant improved beliefs about the self, especially improved self-esteem that is contingent on intrinsic factors such as personal growth and being true to oneself (Nagpaul & Pang, 2017), may lead the youth to strive more for intrinsic goals in early adulthood. As developmental tasks in early adulthood often involve domains such as careers, financial autonomy, and committed relationships, focusing on intrinsic goals in these domains may be rather challenging for them, likely leading to more decreases in life satisfaction over time. As an indirect support, life satisfaction decreases in early adulthood ($\beta = -.44$, $p < .001$), and the association of self-esteem at age 16 with the slope of life satisfaction in early adulthood was identified as negative although not significant[4] ($\beta = -.13$, $p = .120$). A different alternative would be relevant to the Mexican-origin youth's family and broader neighborhood/community backgrounds. Youth with more intrinsic, self-oriented goals may appear not conforming to their socio-oriented family and broader neighborhood/community, which might be a source of stress, inducing decreases in life satisfaction over time. The detected weak moderation of familism ($\beta = .27$, $p = .038$; Supplemental Table S5) suggests that when parent familism is stronger, implicating a stronger cohesiveness with family or macro environment, the prospective association of decreased MV with decreased life satisfaction appears stronger. Future research is needed for testing these alternatives.

We did find a main effect of familism on life satisfaction; specifically, greater increases in familism from age 10 to 16 predicted higher levels of life satisfaction at age 17. Results showed that a more rapid increase in familism from age 10 to 16 predicted higher levels of life satisfaction at age 17 ($\beta = .36$, $p < .001$; see Supplemental Table S4). This is consistent with previous research (e.g., Fuligni et al., 1999; Padilla-Walker et al., 2020) demonstrating that familism has a positive impact on youth well-being, perhaps because familism values promote family social support, contribute to close family relationships, and foster the development of prosocial tendencies among Mexican American youth.

Contrary to H11 and H12, gender and household income did not moderate the impact of MV on life satisfaction. Nativity was also not a moderating factor. Considering the dearth of relevant theory and research available in the literature, it is difficult to speculate about why the predicted moderating effects did not emerge.

Strengths and Limitations

The present study has several strengths. First, this study is among the first few to examine adolescent MV trajectory with longitudinal data. Findings demonstrate that Mexican-origin youth's MV, values which are closely involved in the establishment of their identity among peers, decrease across adolescence. With longitudinal and simultaneously measured data, this study furthermore contributes to the literature by demonstrating the independent developmental courses between MV and familism for Mexican-origin adolescents. As Mexican-origin adolescents often navigate between Hispanic ethnic identity and their American identity, these findings provide us with insights into identity developments in domains such as MV and familism across adolescence.

Next, this study contributes to the literature by identifying antecedent factors of the MV trajectory for Mexican-origin adolescents. Youth's household income is found as an influential

[4] At age 16, youth completed the Self-Description Questionnaire II–Short (SDQII-S; Byrne, 1996; Marsh et al., 2005). Self-esteem was measured by the six-item subscale assessing adolescents' "perceptions of themselves as effective, capable, individuals who are proud and satisfied with the way they are" (Marsh et al., 2005, p. 102). A sample item is "Overall, you have a lot to be proud of." A latent variable of self-esteem was created with the six items. In the latent growth curve model of life satisfaction, the level and the slope were regressed on self-esteem. Self-esteem at age 16 was positively associated with life satisfaction at age 17 ($\beta = .44$, $B = .50$, $SE = .07$, $p < .001$). The association of self-esteem at age 16 with changes in life satisfaction from age 17 to 23 was identified as negative although not significant ($\beta = -.13$, $B = -.14$, $SE = .09$, $p = .120$).

factor on youth's initial MV level. In particular, this study focuses on influences of parent characteristics, including maternal and paternal MV, familism, and education, which have received less empirical attention as compared to peer characteristics. Findings enrich our understanding by indicating that youth raised in families with more positive family environments such as higher household income, higher maternal and paternal education, and lower maternal and paternal MV tend to be less materialistic; higher maternal education is associated with smaller decreases in adolescent MV.

Furthermore, this study addresses prospective associations of the MV developmental trajectory across adolescence with the life satisfaction trajectory later. Note that prior research examining how valuing materialistic goals relates to life satisfaction is based almost exclusively on cross-sectional design, in which participants completed measures of the two constructs at only 1 point in time. To help redress this limitation, we used longitudinal data spanning 14 years to examine how MV from age 10 to 16 was associated with life satisfaction from age 17 to 23. Our findings demonstrate a strong association of larger decreases in MV with higher initial levels of life satisfaction in early adulthood. A relatively weak association of more decreases in MV in adolescence is identified with more decreases in life satisfaction in early adulthood, which suggests alternative mechanisms for prospective associations between MV and life satisfaction.

Finally, the present study focuses on Mexican-origin youth, a population that has generally been neglected in psychological research. Investigation of value adaptation in relation to youth mental health and subjective well-being in a minoritized community is an important endeavor (García Coll et al., 2000). The findings regarding Mexican-origin youth's MV development, the codevelopment of MV and familism, antecedents of MV development, and prospective associations with life satisfaction provide insights into the developmental and social adaptation processes of this underexamined sample.

The present study has several limitations that should be addressed in future research. First, we used a measure of global life satisfaction, but life satisfaction is a multifaceted construct comprised of satisfaction in a variety of specific life domains (e.g., job, finance, health). Future research should explore how MV during adolescence are related to satisfaction in specific life domains that are particularly salient in young adulthood (e.g., job satisfaction, relationship satisfaction). Second, we focused on how MV during adolescence were associated with life satisfaction at a later developmental stage, but materialism values and life satisfaction are both dynamic developmental constructs that are likely to undergo changes over the course of adolescence. Future research should examine codevelopmental associations between these constructs assessed repeatedly at the same ages during adolescence and use a variety of cross-lagged models (Orth et al., 2021) to examine between- and within-person reciprocal transactional associations between MV and life satisfaction. Third, although our focus on a specific ethnic minority community is a strength of the research, future research should explore these developmental processes in youth of different ethnic, racial, and cultural backgrounds to evaluate the generalizability of our findings.

Conclusion

Mean-level changes in MV among Mexican-origin adolescents appear especially important, as Latinxs are a large and rapidly growing ethnic group that continues to experience lower socioeconomic status than Americans of other ethnic backgrounds. Mexican-origin adolescents decreased their endorsement of MV from early to late adolescence. Greater decreases in MV and greater increases in familism across adolescence were associated with higher initial life satisfaction in early adulthood. However, the buffering effect of familism on prospective associations of MV with life satisfaction was not detected. Individuals differed around the normative trend with more positive family environments such as higher household income, higher maternal and paternal education, and lower maternal and paternal MV associated with lower initial MV in adolescence. With these rich longitudinal data, this study contributes to the literature by identifying the developmental trajectory of MV across adolescence and showing how changes in the endorsement of MV across adolescence were prospectively associated with life satisfaction in early adulthood.

References

Ahuvia, A. C., & Wong, N. Y. (2002). Personality and values based materialism: Their relationship and origins. *Journal of Consumer Psychology*, *12*(4), 389–402. https://doi.org/10.1016/S1057-7408(16)30089-4

Allison, P. D. (2003). Missing data techniques for structural equation modeling. *Journal of Abnormal Psychology*, *112*(4), 545–557. https://doi.org/10.1037/0021-843X.112.4.545

Amiot, C. E., de la Sablonnière, R., Terry, D. J., & Smith, J. R. (2007). Integration of social identities in the self: Toward a cognitive-developmental model. *Personality and Social Psychology Review*, *11*(4), 364–388. https://doi.org/10.1177/1088868307304091

Baker, A. M., Moschis, G. P., Ong, F. S., & Pattanapanyasat, R. P. (2013). Materialism and life satisfaction: The role of stress and religiosity. *The Journal of Consumer Affairs*, *47*(3), 548–563. https://doi.org/10.1111/joca.12013

Banerjee, R., & Dittmar, H. (2008). Individual differences in children's materialism: The role of peer relations. *Personality and Social Psychology Bulletin*, *34*(1), 17–31. https://doi.org/10.1177/0146167207309196

Barger, S. D., Donoho, C. J., & Wayment, H. A. (2009). The relative contributions of race/ethnicity, socioeconomic status, health, and social relationships to life satisfaction in the United States. *Quality of Life Research*, *18*(2), 179–189. https://doi.org/10.1007/s11136-008-9426-2

Barnfield, C. (2020). *Materialism and well-being in adolescents* [Doctoral dissertation, University of Surrey]. https://doi.org/10.15126/thesis.00852519

Bauer, M. A., Wilkie, J. E., Kim, J. K., & Bodenhausen, G. V. (2012). Cuing consumerism: Situational materialism undermines personal and social well-being. *Psychological Science*, *23*(5), 517–523. https://doi.org/10.1177/0956797611429579

Behrendt, S. (2014). *lm.beta: Add standardized regression coefficients to linear-model-objects* (R package Version 1.5-1) [Computer software]. https://cran.r-project.org/package=lm.beta

Benenson, J. F. (1993). Greater preference among females than males for dyadic interaction in early childhood. *Child Development*, *64*(2), 544–555. https://doi.org/10.2307/1131268

Beutel, A. M., & Marini, M. M. (1995). Gender and values. *American Sociological Review*, *60*(3), 436–448. https://doi.org/10.2307/2096423

Bradshaw, E. L., Conigrave, J. H., Steward, B. A., Ferber, K. A., Parker, P. D., & Ryan, R. M. (2023). A meta-analysis of the dark side of the American dream: Evidence for the universal wellness costs of prioritizing extrinsic over intrinsic goals. *Journal of Personality and Social Psychology*, *124*(4), 873–899. https://doi.org/10.1037/pspp0000431

Buijzen, M., & Valkenburg, P. M. (2005). Parental mediation of undesired advertising effects. *Journal of Broadcasting & Electronic Media*, *49*(2), 153–165. https://doi.org/10.1207/s15506878jobem4902_1

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Burke, R. J., & Weir, T. (1979). Helping responses of parents and peers and adolescent well-being. *The Journal of Psychology*, *102*(1), 49–62. https://doi.org/10.1080/00223980.1979.9915094

Burroughs, J. E., & Rindfleisch, A. (2002). Materialism and well-being: A conflicting values perspective. *The Journal of Consumer Research*, *29*(3), 348–370. https://doi.org/10.1086/344429

Butkovic, A., Tomas, J., Spanic, A. M., Vukasovic Hlupic, T., & Bratko, D. (2020). Emerging adults versus middle-aged adults: Do they differ in psychological needs, self-esteem and life satisfaction. *Journal of Happiness Studies*, *21*(3), 779–798. https://doi.org/10.1007/s10902-019-00106-w

Buunk, A. P., & Gibbons, F. X. (2007). Social comparison: The end of a theory and the emergence of a field. *Organizational Behavior and Human Decision Processes*, *102*(1), 3–21. https://doi.org/10.1016/j.obhdp.2006.09.007

Byrne, B. M. (1996). Academic self-concept: Its structure, measurement, and relation to academic achievement. In B. A. Bracken (Ed.), *Handbook of self-concept: Developmental, social, and clinical considerations* (pp. 287–316). Wiley.

Campbell, A., Converse, P. E., & Rodgers, W. L. (1976). *The quality of American life: Perceptions, evaluations, and satisfactions*. Russell Sage Foundation.

Carp, F. M., & Carp, A. (1982). Test of a model of domain satisfactions and well-being: Equity considerations. *Research on Aging*, *4*(4), 503–522. https://doi.org/10.1177/0164027582004004005

Carr, A., & Pike, A. (2012). Maternal scaffolding behavior: Links with parenting style and maternal education. *Developmental Psychology*, *48*(2), 543–551. https://doi.org/10.1037/a0025888

Chan, K., & Prendergast, G. (2007). Materialism and social comparison among adolescents. *Social Behavior and Personality*, *35*(2), 213–228. https://doi.org/10.2224/sbp.2007.35.2.213

Chang, L., & Arkin, R. M. (2002). Materialism as an attempt to cope with uncertainty. *Psychology and Marketing*, *19*(5), 389–406. https://doi.org/10.1002/mar.10016

Chaplin, L. N., Hill, R. P., & John, D. R. (2014). Poverty and materialism: A look at impoverished versus affluent children. *Journal of Public Policy & Marketing*, *33*(1), 78–92. https://doi.org/10.1509/jppm.13.050

Chaplin, L. N., & John, D. R. (2005). The development of self-brand connections in children and adolescents. *The Journal of Consumer Research*, *32*(1), 119–129. https://doi.org/10.1086/426622

Chaplin, L. N., & John, D. R. (2007). Growing up in a material world: Age differences in materialism in children and adolescents. *The Journal of Consumer Research*, *34*(4), 480–493. https://doi.org/10.1086/518546

Chaplin, L. N., & John, D. R. (2010). Interpersonal influences on adolescent materialism: A new look at the role of parents and peers. *Journal of Consumer Psychology*, *20*(2), 176–184. https://doi.org/10.1016/j.jcps.2010.02.002

Chen, F. F. (2007). Sensitivity of goodness of fit indexes to lack of measurement invariance. *Structural Equation Modeling*, *14*(3), 464–504. https://doi.org/10.1080/10705510701301834

Cheung, G. W., & Rensvold, R. B. (2002). Evaluating goodness-of-fit indexes for testing measurement invariance. *Structural Equation Modeling*, *9*(2), 233–255. https://doi.org/10.1207/S15328007SEM0902_5

Cohen, P., & Cohen, J. (1996). *Life values and adolescent mental health*. Lawrence Erlbaum.

de Matos, C. A., Vieira, V., Bonfanti, K., & Mette, F. M. B. (2019). Antecedents of indebtedness for low-income consumers: The mediating role of materialism. *Journal of Consumer Marketing*, *36*(1), 92–101. https://doi.org/10.1108/JCM-09-2017-2352

Diener, E., Emmons, R. A., Larsen, R. J., & Griffin, S. (1985). The satisfaction with life scale. *Journal of Personality Assessment*, *49*(1), 71–75. https://doi.org/10.1207/s15327752jpa4901_13

Dittmar, H. (2008). *Consumer culture, identity and well-being: The search for the "good life" and the "body perfect"*. Psychology Press.

Dittmar, H., Bond, R., Hurst, M., & Kasser, T. (2014). The relationship between materialism and personal well-being: A meta-analysis. *Journal of Personality and Social Psychology*, *107*(5), 879–924. https://doi.org/10.1037/a0037409

Donnelly, G. E., Iyer, R., & Howell, R. T. (2012). The Big Five personality traits, material values, and financial well-being of self-described money managers. *Journal of Economic Psychology*, *33*(6), 1129–1142. https://doi.org/10.1016/j.joep.2012.08.001

Donnelly, G. E., Ksendzova, M., Howell, R. T., Vohs, K. D., & Baumeister, R. F. (2016). Buying to blunt negative feelings: Materialistic escape from the self. *Review of General Psychology*, *20*(3), 272–316. https://doi.org/10.1037/gpr0000078

Eccles, J. S., Wigfield, A., Flanagan, C. A., Miller, C., Reuman, D. A., & Yee, D. (1989). Self-concepts, domain values, and self-esteem: Relations and changes at early adolescence. *Journal of Personality*, *57*(2), 283–310. https://doi.org/10.1111/j.1467-6494.1989.tb00484.x

Erikson, E. H. (1968). *Identity: Youth and crisis*. W.W. Norton.

Erikson, E. H. (1994). *Identity and the life cycle*. W.W. Norton.

Felix, R., Ahmed, Z. U., & Hinck, W. (2013). Gender issues in consumer materialism: The case of Mexico. *Journal of Transnational Management*, *18*(2), 82–100. https://doi.org/10.1080/15475778.2013.782230

Fischer, K. W. (1980). A theory of cognitive development: The control and construction of hierarchies of skills. *Psychological Review*, *87*(6), 477–531. https://doi.org/10.1037/0033-295X.87.6.477

Flouri, E. (2004). Exploring the relationship between mothers' and fathers' parenting practices and children's materialist values. *Journal of Economic Psychology*, *25*(6), 743–752. https://doi.org/10.1016/j.joep.2003.06.005

Flurry, L. A., & Swimberghe, K. R. (2021). The affluenza epidemic: Consequences of parent–child value congruence in a material world. *Journal of Consumer Marketing*, *38*(2), 201–210. https://doi.org/10.1108/JCM-04-2019-3159

Frost, R. O., Kyrios, M., McCarthy, K. D., & Matthews, Y. (2007). Self-ambivalence and attachment to possessions. *Journal of Cognitive Psychotherapy*, *21*(3), 232–242. https://doi.org/10.1891/088983907781494582

Fuligni, A. J., Tseng, V., & Lam, M. (1999). Attitudes toward family obligations among American adolescents with Asian, Latin American, and European backgrounds. *Child Development*, *70*(4), 1030–1044. https://doi.org/10.1111/1467-8624.00075

García Coll, C. G., Akerman, A., & Cicchetti, D. (2000). Cultural influences on developmental processes and outcomes: Implications for the study of development and psychopathology. *Development and Psychopathology*, *12*(3), 333–356. https://doi.org/10.1017/S0954579400003059

Ger, G., & Belk, R. W. (1999). Accounting for materialism in four cultures. *Journal of Material Culture*, *4*(2), 183–204. https://doi.org/10.1177/135918359900400204

Goldberg, M. E., Gorn, G. J., Peracchio, L. A., & Bamossy, G. (2003). Understanding materialism among youth. *Journal of Consumer Psychology*, *13*(3), 278–288. https://doi.org/10.1207/S15327663JCP1303_09

Granott, N., Fischer, K. W., & Parziale, J. (2002). Bridging to the unknown: A transition mechanism in learning and development. In N. Granott & J. Parziale (Eds.), *Microdevelopment: Transition processes in development and learning* (pp. 131–156). Cambridge University Press. https://doi.org/10.1017/CBO9780511489709.006

Greenberg, M. T., Siegel, J. M., & Leitch, C. J. (1983). The nature and importance of attachment relationships to parents and peers during adolescence. *Journal of Youth and Adolescence*, *12*(5), 373–386. https://doi.org/10.1007/BF02088721

Grotevant, H. D. (1987). Toward a process model of identity formation. *Journal of Adolescent Research*, *2*(3), 203–222. https://doi.org/10.1177/074355488723003

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This article is intended solely for the personal use of the individual user and is not to be disseminated broadly.

Grougiou, V., & Moschis, G. P. (2015). Antecedents of young adults' materialistic values. *Journal of Consumer Behaviour*, *14*(2), 115–126. https://doi.org/10.1002/cb.1505

Harter, S. (1999). *The construction of the self: A developmental perspective*. Guilford Press.

Heaney, J. G., Goldsmith, R. E., & Jusoh, W. J. W. (2005). Status consumption among Malaysian consumers: Exploring its relationships with materialism and attention-to-social-comparison-information. *Journal of International Consumer Marketing*, *17*(4), 83–98. https://doi.org/10.1300/J046v17n04_05

Hu, L. T., & Bentler, P. M. (1999). Cutoff criteria for fit indexes in covariance structure analysis: Conventional criteria versus new alternatives. *Structural Equation Modeling*, *6*(1), 1–55. https://doi.org/10.1080/10705519909540118

Islam, T., Wei, J., Sheikh, Z., Hameed, Z., & Azam, R. I. (2017). Determinants of compulsive buying behavior among young adults: The mediating role of materialism. *Journal of Adolescence*, *61*(1), 117–130. https://doi.org/10.1016/j.adolescence.2017.10.004

John, D. R. (1999). Consumer socialization of children: A retrospective look at twenty-five years of research. *The Journal of Consumer Research*, *26*(3), 183–213. https://doi.org/10.1086/209559

Jorgensen, T. D., Pornprasertmanit, S., Schoemann, A. M., & Rosseel, Y. (2022). *semTools: Useful tools for structural equation modeling* (Version 0.5-6) [Computer software]. https://CRAN.R-project.org/package=semTools

Kamineni, R. (2005). Influence of materialism, gender and nationality on consumer brand perceptions. *Journal of Targeting, Measurement and Analysis for Marketing*, *14*(1), 25–32. https://doi.org/10.1057/palgrave.jt.5740167

Kasser, T. (2005). Frugality, generosity, and materialism in children and adolescents. In K. A. Moore & L. H. Lippman (Eds.), *What do children need to flourish? Conceptualizing and measuring indicators of positive development* (pp. 357–373). Springer. https://doi.org/10.1007/0-387-23823-9_22

Kasser, T., & Ahuvia, A. (2002). Materialistic values and well-being in business students. *European Journal of Social Psychology*, *32*(1), 137–146. https://doi.org/10.1002/ejsp.85

Kasser, T., Rosenblum, K. L., Sameroff, A. J., Deci, E. L., Niemiec, C. P., Ryan, R. M., & Hawks, S. (2014). Changes in materialism, changes in psychological well-being: Evidence from three longitudinal studies and an intervention experiment. *Motivation and Emotion*, *38*(1), 1–22. https://doi.org/10.1007/s11031-013-9371-4

Knight, G. P., Berkel, C., Umaña-Taylor, A. J., Gonzales, N. A., Ettekal, I., Jaconis, M., & Boyd, B. M. (2011). The familial socialization of culturally related values in Mexican American families. *Journal of Marriage and Family*, *73*(5), 913–925. https://doi.org/10.1111/j.1741-3737.2011.00856.x

Ku, L. (2015). Development of materialism in adolescence: The longitudinal role of life satisfaction among Chinese youths. *Social Indicators Research*, *124*(1), 231–247. https://doi.org/10.1007/s11205-014-0787-3

Kuanr, A., Pradhan, D., & Chaudhuri, H. R. (2020). I (do not) consume; therefore, I am: Investigating material-ism and voluntary simplicity through a moderated mediation model. *Psychology and Marketing*, *37*(2), 260–277. https://doi.org/10.1002/mar.21305

Kuster, F., & Orth, U. (2013). The long-term stability of self-esteem: Its time-dependent decay and nonzero asymptote. *Personality and Social Psychology Bulletin*, *39*(5), 677–690. https://doi.org/10.1177/0146167213480189

Lee, M. S., & Ahn, C. S. Y. (2016). Anti-consumption, materialism, and consumer well-being. *The Journal of Consumer Affairs*, *50*(1), 18–47. https://doi.org/10.1111/joca.12089

Mangleburg, T. F., Grewal, D., & Bristol, T. (1997). Socialization, gender, and adolescent's self-reports of their generalized use of product labels. *The Journal of Consumer Affairs*, *31*(2), 255–279. https://doi.org/10.1111/j.1745-6606.1997.tb00391.x

Manzi, C., Vignoles, V. L., Regalia, C., & Scabini, E. (2006). Cohesion and enmeshment revisited: Differentiation, identity, and well-being in two European cultures. *Journal of Marriage and Family*, *68*(3), 673–689. https://doi.org/10.1111/j.1741-3737.2006.00282.x

Marsh, H. W., Trautwein, U., Lüdtke, O., Köller, O., & Baumert, J. (2005). Academic self-concept, interest, grades, and standardized test scores: Reciprocal effects models of causal ordering. *Child Development*, *76*(2), 397–416. https://doi.org/10.1111/j.1467-8624.2005.00853.x

Meade, A. W., Johnson, E. C., & Braddy, P. W. (2006). The utility of alternative fit indices in tests of measurement invariance. *Academy of Management Proceedings*, *2006*(1), B1–B6. https://doi.org/10.5465/ambpp.2006.27182124

Miller, S. A. (1998). *Developmental research methods* (2nd ed.). Prentice-Hall.

Moldes, O., & Ku, L. (2020). Materialistic cues make us miserable: A meta-analysis of the experimental evidence for the effects of materialism on individual and societal well-being. *Psychology and Marketing*, *37*(10), 1396–1419. https://doi.org/10.1002/mar.21387

Moschis, G. P., & Churchill, G. A., Jr. (1978). Consumer socialization: A theoretical and empirical analysis. *Journal of Marketing Research*, *15*(4), 599–609. https://doi.org/10.1177/002224377801500409

Nagpaul, T., & Pang, J. S. (2017). Extrinsic and intrinsic contingent self-esteem and materialism: A correlational and experimental investigation. *Psychology and Marketing*, *34*(6), 610–622. https://doi.org/10.1002/mar.21009

Neitzel, C., & Stright, D. A. (2004). Parenting behaviours during child problem solving: The roles of child temperament, mother education and personality, and the problem-solving context. *International Journal of Behavioral Development*, *28*(2), 166–179. https://doi.org/10.1080/01650250344000370

Nickerson, C., Schwarz, N., Diener, E., & Kahneman, D. (2003). Zeroing in on the dark side of the American Dream: A closer look at the negative consequences of the goal for financial success. *Psychological Science*, *14*(6), 531–536. https://doi.org/10.1046/j.0956-7976.2003.psci_1461.x

Noguti, V., & Bokeyar, A. L. (2014). Who am I? The relationship between self-concept uncertainty and materialism. *International Journal of Psychology*, *49*(5), 323–333. https://doi.org/10.1002/ijop.12031

Orth, U., Clark, D. A., Donnellan, M. B., & Robins, R. W. (2021). Testing prospective effects in longitudinal research: Comparing seven competing cross-lagged models. *Journal of Personality and Social Psychology*, *120*(4), 1013–1034. https://doi.org/10.1037/pspp0000358

Padilla-Walker, L. M., Millett, M. A., & Memmott-Elison, M. K. (2020). Can helping others strengthen teens? Character strengths as mediators between prosocial behavior and adolescents' internalizing symptoms. *Journal of Adolescence*, *79*(1), 70–80. https://doi.org/10.1016/j.adolescence.2020.01.001

Padilla-Walker, L. M., Nielson, M. G., & Day, R. D. (2016). The role of parental warmth and hostility on adolescents' prosocial behavior toward multiple targets. *Journal of Family Psychology*, *30*(3), 331–340. https://doi.org/10.1037/fam0000157

Park, J. K., & John, D. R. (2011). More than meets the eye: The influence of implicit and explicit self-esteem on materialism. *Journal of Consumer Psychology*, *21*(1), 73–87. https://doi.org/10.1016/j.jcps.2010.09.001

Perez-Brena, N. J., Updegraff, K. A., & Umaña-Taylor, A. J. (2012). Father- and mother–adolescent decision-making in Mexican-origin families. *Journal of Youth and Adolescence*, *41*(4), 460–473. https://doi.org/10.1007/s10964-011-9660-8

Pinquart, M. (2017). Associations of parenting dimensions and styles with externalizing problems of children and adolescents: An updated meta-analysis. *Developmental Psychology*, *53*(5), 873–932. https://doi.org/10.1037/dev0000295

Podoshen, J. S., & Andrzejewski, S. A. (2012). An examination of the relationships between materialism, conspicuous consumption, impulse buying, and brand loyalty. *Journal of Marketing Theory and Practice*, *20*(3), 319–334. https://doi.org/10.2753/MTP1069-6679200306

This document is copyrighted by the American Psychological Association or one of its allied publishers.
This article is intended solely for the personal use of the individual user and is not to be disseminated broadly.

Reeves, R. A., Baker, G. A., & Truluck, C. S. (2012). Celebrity worship, materialism, compulsive buying, and the empty self. *Psychology and Marketing*, *29*(9), 674–679. https://doi.org/10.1002/mar.20553

Retelsdorf, J., Schwartz, K., & Asbrock, F. (2015). "Michael can't read!" Teachers' gender stereotypes and boys' reading self-concept. *Journal of Educational Psychology*, *107*(1), 186–194. https://doi.org/10.1037/a0037107

Revelle, W. (2020). *psych: Procedures for psychological, psychometric, and personality research* (R package Version 2.1.6) [Computer software]. Northwestern University. https://CRAN.R-project.org/package=psych

Richins, M. L. (2011). Materialism, transformation expectations, and spending: Implications for credit use. *Journal of Public Policy & Marketing*, *30*(2), 141–156. https://doi.org/10.1509/jppm.30.2.141

Richins, M. L. (2017). Materialism pathways: The processes that create and perpetuate materialism. *Journal of Consumer Psychology*, *27*(4), 480–499. https://doi.org/10.1016/j.jcps.2017.07.006

Richins, M. L., & Dawson, S. (1992). A consumer values orientation and its measurement: Scale development and validation. *The Journal of Consumer Research*, *19*(3), 303–317. https://doi.org/10.1086/209304

Rindfleisch, A., Burroughs, J. E., & Denton, F. (1997). Family structure, materialism, and compulsive consumption. *The Journal of Consumer Research*, *23*(4), 312–325. https://doi.org/10.1086/209486

Rindfleisch, A., Burroughs, J. E., & Wong, N. (2009). The safety of objects: Materialism, existential insecurity, and brand connection. *The Journal of Consumer Research*, *36*(1), 1–16. https://doi.org/10.1086/595718

Roberts, J. A., Tanner, J. F., Jr., & Manolis, C. (2005). Materialism and the family structure–stress relation. *Journal of Consumer Psychology*, *15*(2), 183–190. https://doi.org/10.1207/s15327663jcp1502_10

Rosenberg, M. (1986). Self-Concept from middle childhood through adolescence. In J. Suls & A. G. Greenwald (Eds.), *Psychological perspectives on the self* (pp. 107–135). Lawrence Erlbaum.

Rosseel, Y. (2012). lavaan: An R package for structural equation modeling. *Journal of Statistical Software*, *48*(2), 1–36. https://doi.org/10.18637/jss.v048.i02

Ruvio, A., Somer, E., & Rindfleisch, A. (2014). When bad gets worse: The amplifying effect of materialism on traumatic stress and maladaptive consumption. *Journal of the Academy of Marketing Science*, *42*(1), 90–101. https://doi.org/10.1007/s11747-013-0345-6

Ryan, L., & Dziurawiec, S. (2001). Materialism and its relationship to life satisfaction. *Social Indicators Research*, *55*(2), 185–197. https://doi.org/10.1023/A:1011002123169

Ryan, R. M., & Deci, E. L. (2000). Self-determination theory and the facilitation of intrinsic motivation, social development, and well-being. *American Psychologist*, *55*(1), 68–78. https://doi.org/10.1037/0003-066X.55.1.68

Sabogal, F., Marín, G., Otero-Sabogal, R., Marín, B. V., & Perez-Stable, E. J. (1987). Hispanic familism and acculturation: What changes and what doesn't? *Hispanic Journal of Behavioral Sciences*, *9*(4), 397–412. https://doi.org/10.1177/07399863870094003

Sankoff, G. (2008). *Cross-sectional and longitudinal studies* (Vol. 2). De Gruyter Mouton.

Schafer, J. L., & Graham, J. W. (2002). Missing data: Our view of the state of the art. *Psychological Methods*, *7*(2), 147–177. https://doi.org/10.1037/1082-989X.7.2.147

Schor, J. (2004). *Born to buy: The commercialized child and the new consumer culture*. Scribner.

Schroeder, J. E., & Dugal, S. S. (1995). Psychological correlates of the materialism construct. *Journal of Social Behavior and Personality*, *10*(1), 243–253.

Segal, B., & Podoshen, J. S. (2013). An examination of materialism, conspicuous consumption and gender differences. *International Journal of Consumer Studies*, *37*(2), 189–198. https://doi.org/10.1111/j.1470-6431.2012.01099.x

Sheldon, K. M., & Kasser, T. (2008). Psychological threat and extrinsic goal striving. *Motivation and Emotion*, *32*(1), 37–45. https://doi.org/10.1007/s11031-008-9081-5

Sheldon, K. M., & Schüler, J. (2011). Wanting, having, and needing: Integrating motive disposition theory and self-determination theory. *Journal of Personality and Social Psychology*, *101*(5), 1106–1123. https://doi.org/10.1037/a0024952

Shi, B., & Xie, H. (2012). Socialization of physical and social aggression in early adolescents' peer groups: High-status peers, gender, and individual status. *Social Development*, *21*(1), 170–194. https://doi.org/10.1111/j.1467-9507.2011.00621.x

Shi, B., Zhang, D., Xie, H., & Zhou, Y. (2016). Antecedents of adolescent Chinese consumers' purchase intention for local brands: The moderating influence of materialistic values. *Journal of Consumer Marketing*, *33*(4), 292–301. https://doi.org/10.1108/JCM-07-2014-1045

Smolak, L., & Stein, J. A. (2006). The relationship of drive for muscularity to sociocultural factors, self-esteem, physical attributes gender role, and social comparison in middle school boys. *Body Image*, *3*(2), 121–129. https://doi.org/10.1016/j.bodyim.2006.03.002

Stein, G. L., Cupito, A. M., Mendez, J. L., Prandoni, J., Huq, N., & Westerberg, D. (2014). Familism through a developmental lens. *Journal of Latina/o Psychology*, *2*(4), 224–250. https://doi.org/10.1037/lat0000025

Stein, J. A., Newbomb, M. D., & Bentler, P. M. (1992). The effect of agency and communality on self-esteem: Gender differences in longitudinal data. *Sex Roles*, *26*(11–12), 465–483. https://doi.org/10.1007/BF00289869

Tajfel, H. (1981). *Human groups and social categories: Studies in social psychology*. Cambridge University Press.

Telzer, E. H. (2011). Expanding the acculturation gap-distress model: An integrative review of research. *Human Development*, *53*(6), 313–340. https://doi.org/10.1159/000322476

Thompson, C. J., & Hirschman, E. C. (1995). Understanding the socialized body: A poststructuralist analysis of consumers' self-conceptions, body images, and self-care practices. *The Journal of Consumer Research*, *22*(2), 139–153. https://doi.org/10.1086/209441

Trim, J. G. (2008). *Materialistic values and depressive symptoms among Mexican-American adults* [Doctoral dissertation, Arizona State University]. ProQuest Dissertations and Theses. https://www.pqdtcn.com/thesisDetails/1B96735E0C862F8E1789D15AF533CA1B

Twenge, J. M., & Campbell, W. K. (2001). Age and birth cohort differences in self-esteem: A cross-temporal meta-analysis. *Personality and Social Psychology Review*, *5*(4), 321–344. https://doi.org/10.1207/S15327957PSPR0504_3

Updegraff, K. A., McHale, S. M., Whiteman, S. D., Thayer, S. M., & Delgado, M. Y. (2005). Adolescent sibling relationships in Mexican American families: Exploring the role of familism. *Journal of Family Psychology*, *19*(4), 512–522. https://doi.org/10.1037/0893-3200.19.4.512

Usmiani, S., & Daniluk, J. (1997). Mothers and their adolescent daughters: Relationship between self-esteem, gender role identity, body image. *Journal of Youth and Adolescence*, *26*(1), 45–62. https://doi.org/10.1023/A:1024588112108

Van den Akker, A. L., Deković, M., Asscher, J., & Prinzie, P. (2014). Mean-level personality development across childhood and adolescence: A temporary defiance of the maturity principle and bidirectional associations with parenting. *Journal of Personality and Social Psychology*, *107*(4), 736–750. https://doi.org/10.1037/a0037248

Vansteenkiste, M., Lens, W., & Deci, E. L. (2006). Intrinsic versus extrinsic goal contents in self-determination theory: Another look at the quality of academic motivation. *Educational Psychologist*, *41*(1), 19–31. https://doi.org/10.1207/s15326985ep4101_4

Watson, J. J. (2003). The relationship of materialism to spending tendencies, saving, and debt. *Journal of Economic Psychology*, *24*(6), 723–739. https://doi.org/10.1016/j.joep.2003.06.001

Wheeler, L. A., Zeiders, K. H., Updegraff, K. A., Umaña-Taylor, A. J., Rodríguez de Jesús, S. A., & Perez-Brena, N. J. (2017). Mexican-origin

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youth's risk behavior from adolescence to young adulthood: The role of familism values. *Developmental Psychology*, *53*(1), 126–137. https://doi.org/10.1037/dev0000251

Wickham, H. (2016). *ggplot2: Elegant graphics for data analysis* (2nd ed.). Springer.

Widaman, K. F., Ferrer, E., & Conger, R. D. (2010). Factorial invariance within longitudinal structural equation models: Measuring the same construct across time. *Child Development Perspectives*, *4*(1), 10–18. https://doi.org/10.1111/j.1750-8606.2009.00110.x

Willroth, E. C., Atherton, O. E., & Robins, R. W. (2021). Life satisfaction trajectories during adolescence and the transition to young adulthood: Findings from a longitudinal study of Mexican-origin youth. *Journal of Personality and Social Psychology*, *120*(1), 192–205. https://doi.org/10.1037/pspp0000294

Wooten, D. B. (2006). From labeling possessions to possessing labels: Ridicule and socialization among adolescents. *The Journal of Consumer Research*, *33*(2), 188–198. https://doi.org/10.1086/506300

Yan, Z., & Fischer, K. (2002). Always under construction: Dynamic variations in adult cognitive micro-development. *Human Development*, *45*(3), 141–160. https://doi.org/10.1159/000057070

Yoon, S., & Kim, H. C. (2016). Keeping the American dream alive: The interactive effect of perceived economic mobility and materialism on impulsive spending. *Journal of Marketing Research*, *53*(5), 759–772. https://doi.org/10.1509/jmr.15.0053

Young, M. H., Miller, B. C., Norton, M. C., & Hill, E. J. (1995). The effect of parental supportive behaviors on life satisfaction of adolescent offspring. *Journal of Marriage and Family*, *57*(3), 813–822. https://doi.org/10.2307/353934

Received January 12, 2023
Revision received September 10, 2023
Accepted September 20, 2023 ■

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